Result Compendium

$Z \to \ell\ell\gamma$ — Unconverted γ

η-only binning, loose isolation

Cell-Energy Reweighting Pipeline

May 3, 2026

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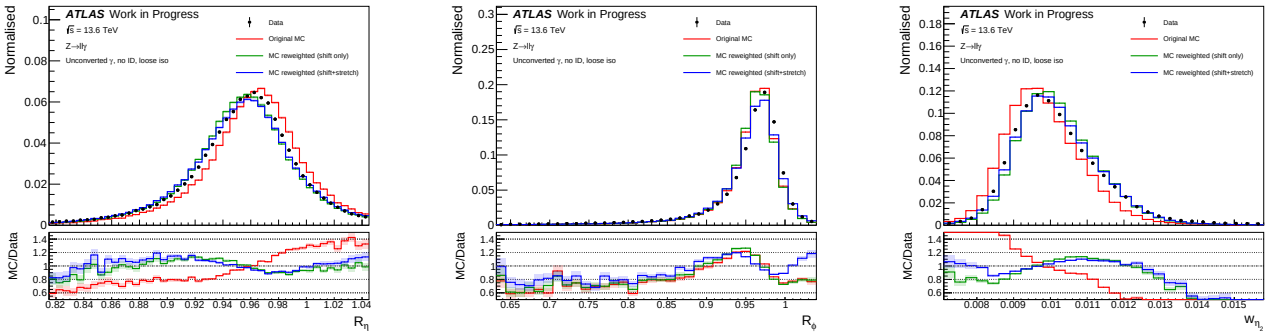
1 Chi-Squared Summary

Table 1: Per-$|\eta|$-bin χ^2/n_{df} for $Z \to \ell\ell\gamma$, Unconverted scenario. Columns show χ^2/n_{df} between each MC variant and data (cell-computed).

$\lvert\eta\rvert$ bin	R_η				R_ϕ				$w_{\eta 2}$			
(range)	MC	M1	M2	Fud	MC	M1	M2	Fud	MC	M1	M2	Fud
[0.00, 0.20)	0.0002	0.0002	0.0002	0.0002	0.0004	0.0005	0.0003	0.0002	0.0008	0.0003	0.0002	0.0003
[0.20, 0.40)	0.0002	0.0001	0.0001	0.0001	0.0003	0.0004	0.0003	0.0002	0.0014	0.0004	0.0003	0.0003
[0.40, 0.60)	0.0003	0.0002	0.0002	0.0002	0.0005	0.0005	0.0003	0.0002	0.0016	0.0003	0.0003	0.0003
[0.60, 0.80)	0.0003	0.0002	0.0002	0.0002	0.0003	0.0004	0.0003	0.0002	0.0016	0.0004	0.0003	0.0003
[0.80, 1.00)	0.0005	0.0002	0.0003	0.0002	0.0007	0.0005	0.0003	0.0003	0.0006	0.0004	0.0002	0.0003
[1.00, 1.20)	0.0004	0.0001	0.0002	0.0002	0.0010	0.0008	0.0004	0.0003	0.0005	0.0006	0.0003	0.0003
[1.20, 1.30)	0.0007	0.0004	0.0005	0.0004	0.0008	0.0010	0.0004	0.0004	0.0010	0.0007	0.0004	0.0004
[1.30, 1.37)	0.0027	0.0011	0.0017	0.0019	0.0014	0.0016	0.0011	0.0014	0.0022	0.0020	0.0019	0.0012
[1.52, 1.60)	0.0014	0.0027	0.0015	0.0008	0.0010	0.0036	0.0011	0.0011	0.0072	0.0014	0.0011	0.0010
[1.60, 1.80)	0.0013	0.0005	0.0004	0.0003	0.0004	0.0011	0.0006	0.0005	0.0080	0.0004	0.0004	0.0003
[1.80, 2.00)	0.0023	0.0005	0.0008	0.0004	0.0005	0.0011	0.0009	0.0004	0.0171	0.0016	0.0008	0.0004
[2.00, 2.20)	0.0034	0.0004	0.0004	0.0006	0.0006	0.0009	0.0005	0.0004	0.0273	0.0005	0.0003	0.0004
[2.20, 2.40)	0.0054	0.0006	0.0007	0.0005	0.0011	0.0014	0.0007	0.0008	0.0296	0.0005	0.0005	0.0005
Average	0.0015	0.0006	0.0005	0.0005	0.0007	0.0011	0.0006	0.0005	0.0076	0.0007	0.0005	0.0005

2 Integrated Shower Shape Comparisons

Data versus uncorrected MC, M1, and M2 corrections (all $|\eta|$ bins combined).



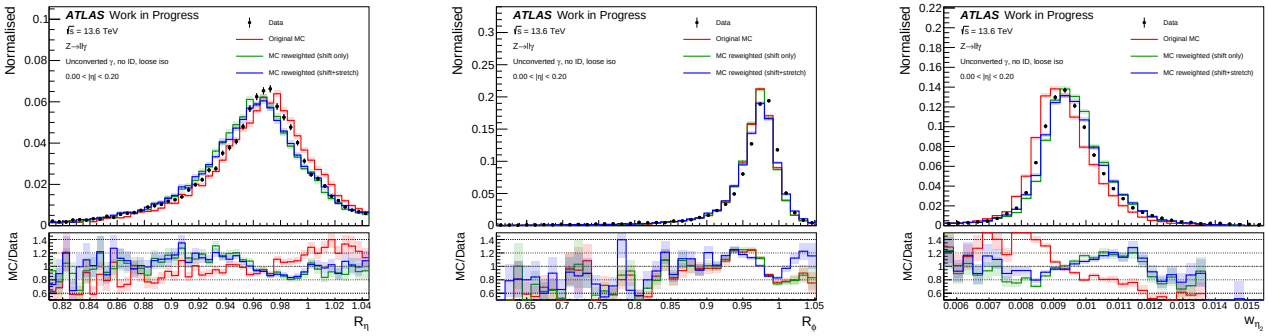
(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 1: Integrated shower shape comparisons: Data vs uncorrected MC, M1 and M2.

3 Per-$|\eta|$ Shower Shape Comparisons

Data versus uncorrected MC, M1, and M2 corrections for each $|\eta|$ bin.

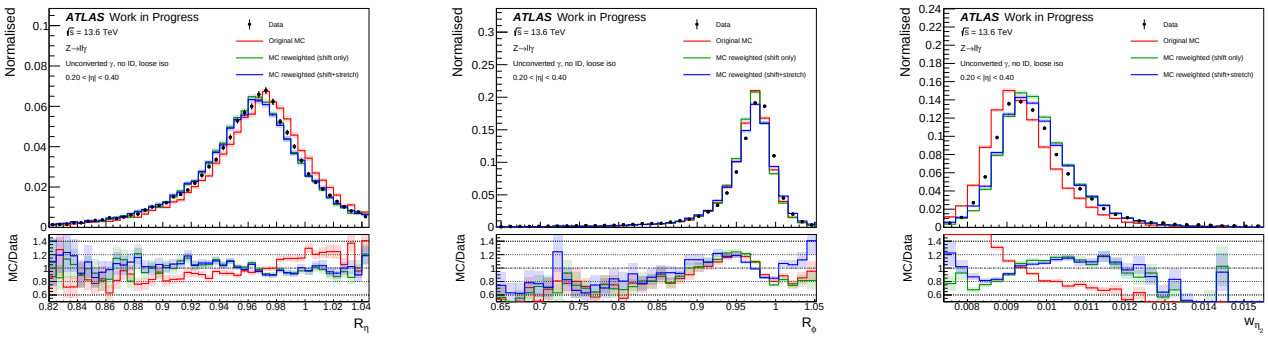
$|\eta| \in [0.00, 0.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 2: Shower shape comparisons, $|\eta| \in [0.00, 0.20)$.

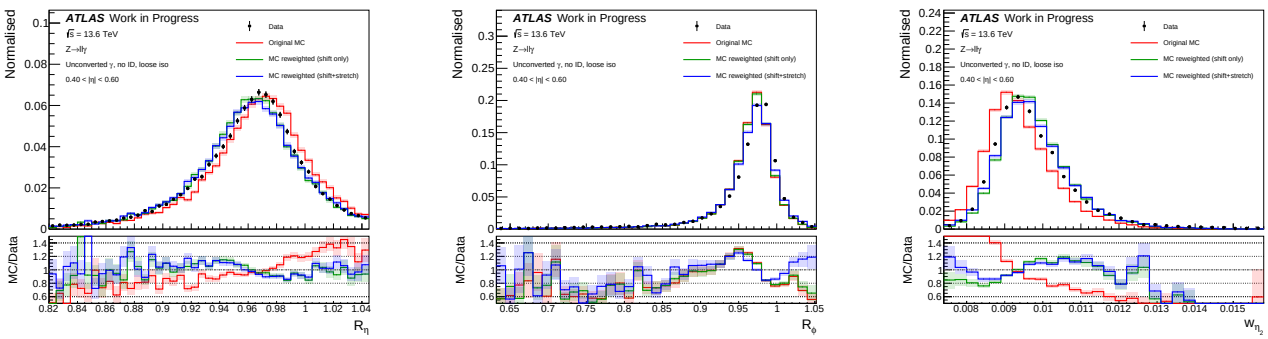
$|\eta| \in [0.20, 0.40)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 3: Shower shape comparisons, $|\eta| \in [0.20, 0.40)$.

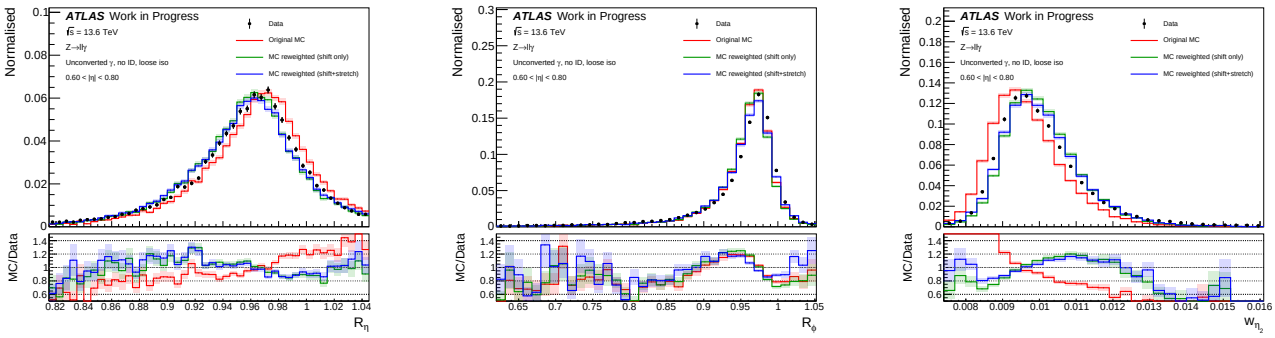
$|\eta| \in [0.40, 0.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 4: Shower shape comparisons, $|\eta| \in [0.40, 0.60)$.

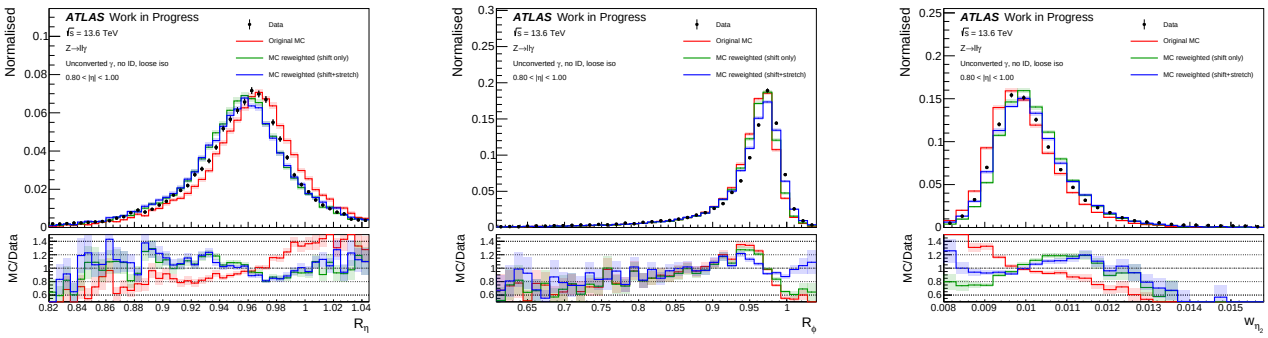
$|\eta| \in [0.60, 0.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 5: Shower shape comparisons, $|\eta| \in [0.60, 0.80)$.

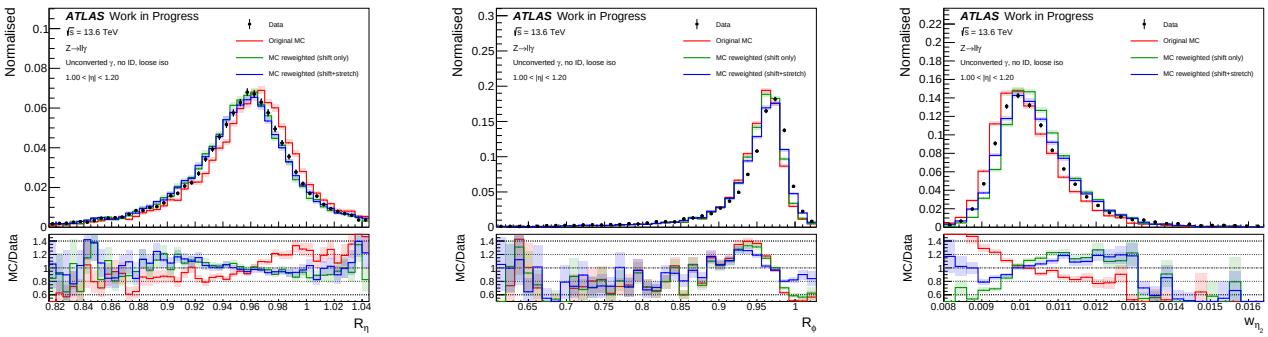
$|\eta| \in [0.80, 1.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 6: Shower shape comparisons, $|\eta| \in [0.80, 1.00)$.

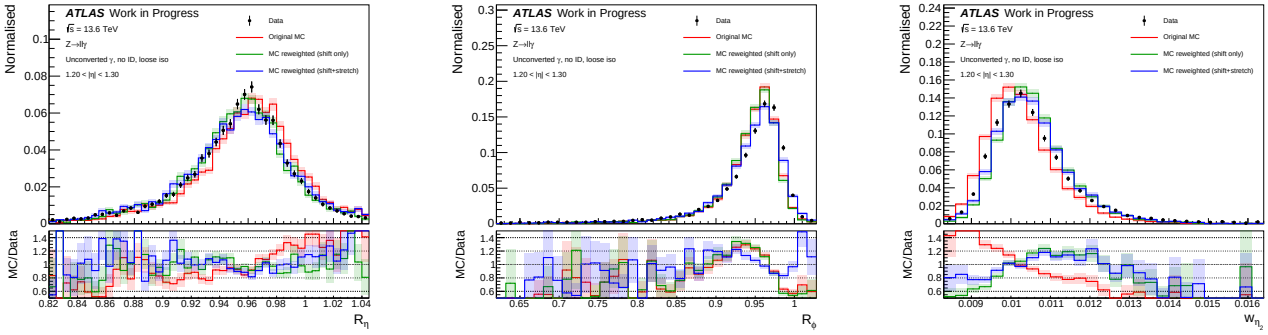
$|\eta| \in [1.00, 1.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 7: Shower shape comparisons, $|\eta| \in [1.00, 1.20)$.

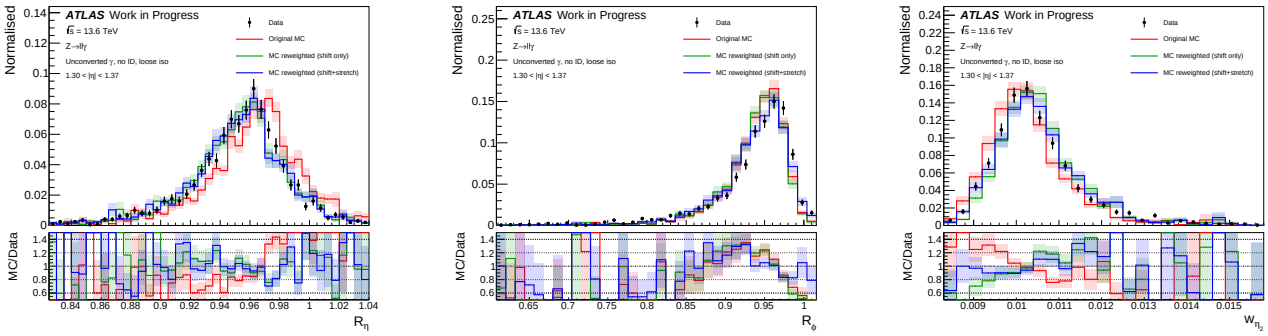
$|\eta| \in [1.20, 1.30)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 8: Shower shape comparisons, $|\eta| \in [1.20, 1.30)$.

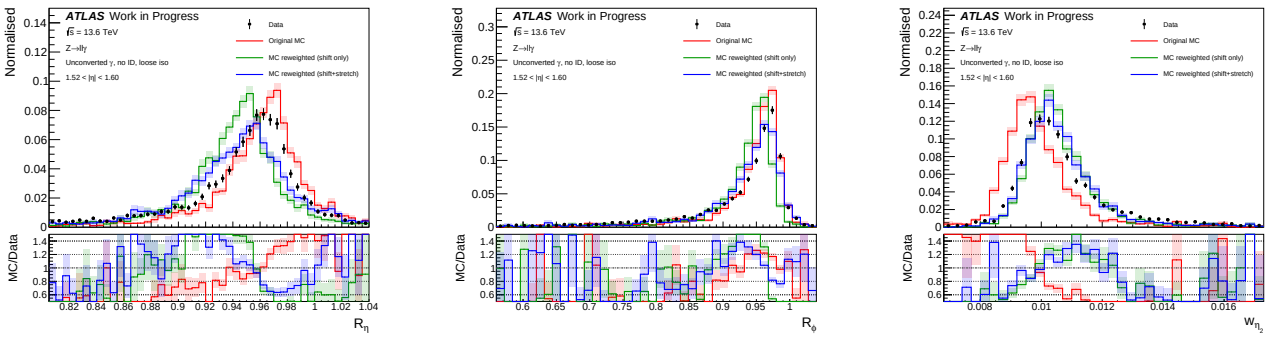
$|\eta| \in [1.30, 1.37)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 9: Shower shape comparisons, $|\eta| \in [1.30, 1.37)$.

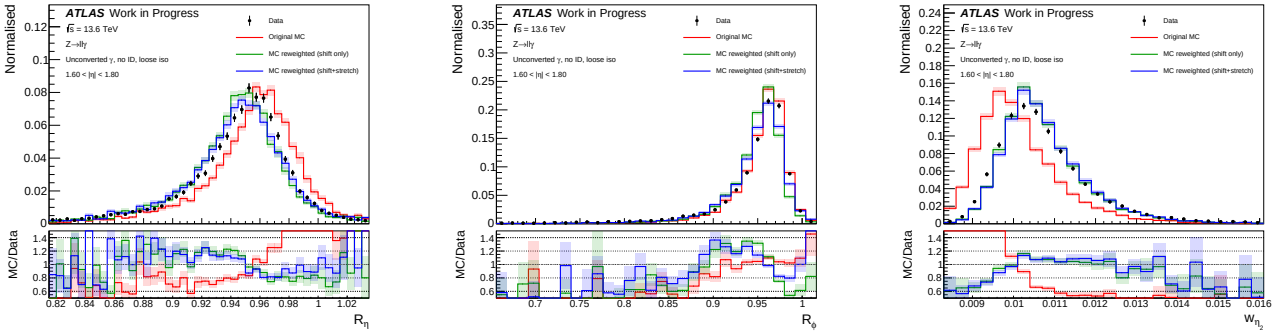
$|\eta| \in [1.52, 1.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 10: Shower shape comparisons, $|\eta| \in [1.52, 1.60)$.

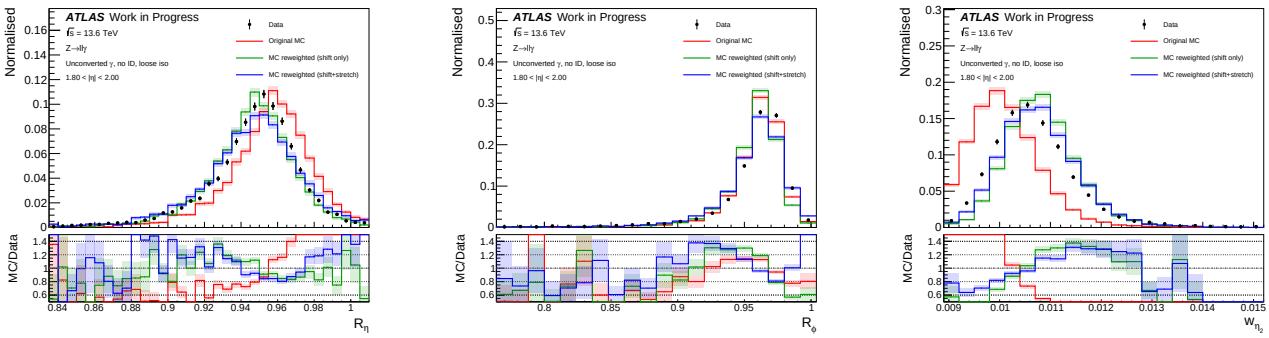
$|\eta| \in [1.60, 1.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 11: Shower shape comparisons, $|\eta| \in [1.60, 1.80)$.

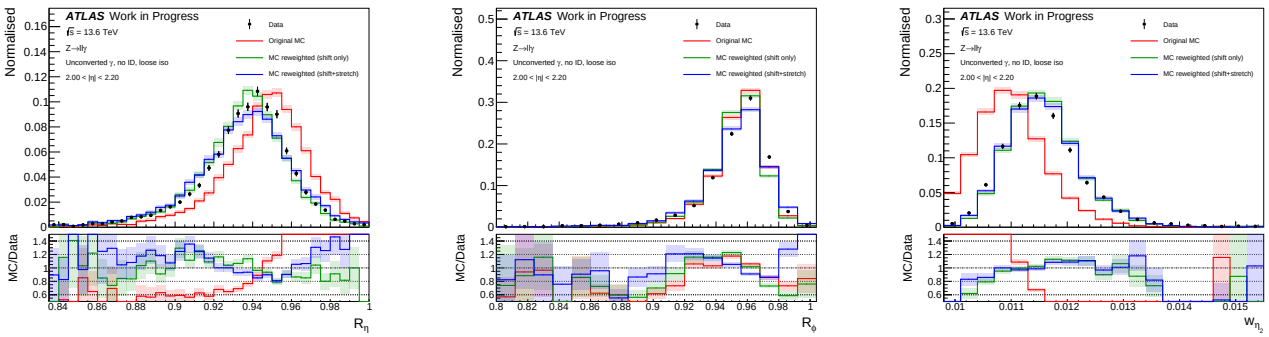
$|\eta| \in [1.80, 2.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 12: Shower shape comparisons, $|\eta| \in [1.80, 2.00)$.

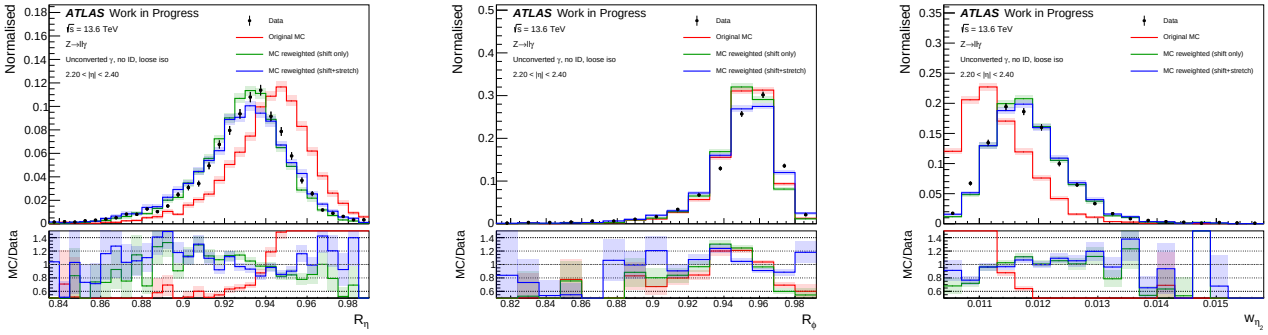
$|\eta| \in [2.00, 2.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 13: Shower shape comparisons, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$



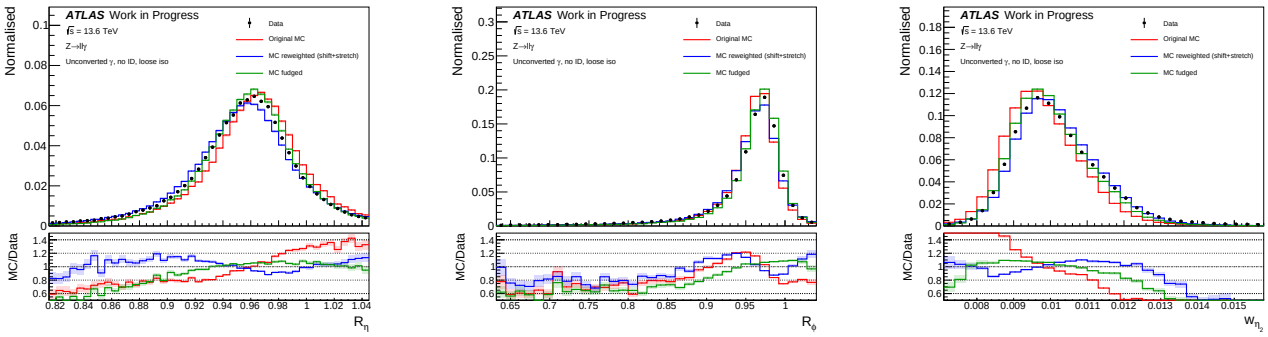
(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 14: Shower shape comparisons, $|\eta| \in [2.20, 2.40)$.

4 Shower Shape Comparisons vs. ATLAS Fudge Factors

Data (cell-computed) versus original MC, M2 cell reweighting, and ATLAS fudge factors (stored-branch fudged MC). Provides a direct comparison of the cell-reweighting correction against the existing ATLAS shower-shape corrections.

Integrated (all $|\eta|$ bins)

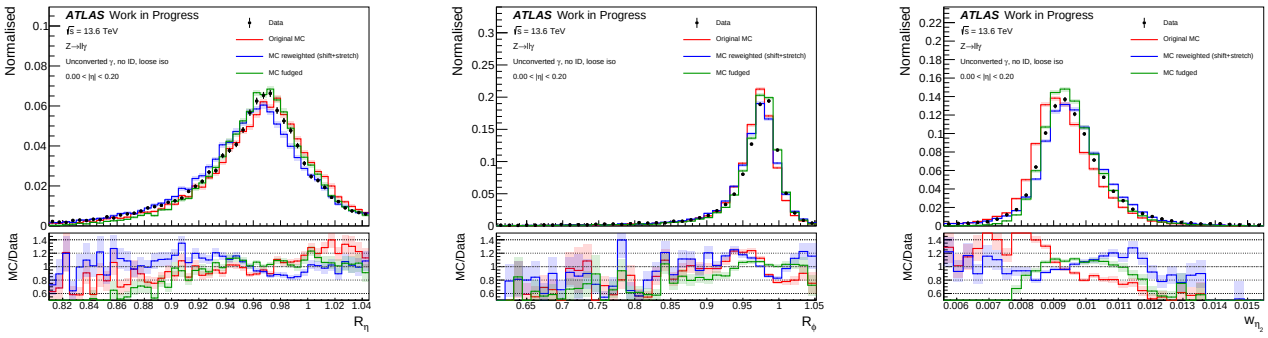


(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 15: Integrated: Data vs. original MC, M2 reweighting, and ATLAS fudged MC.

Per $|\eta|$ bin

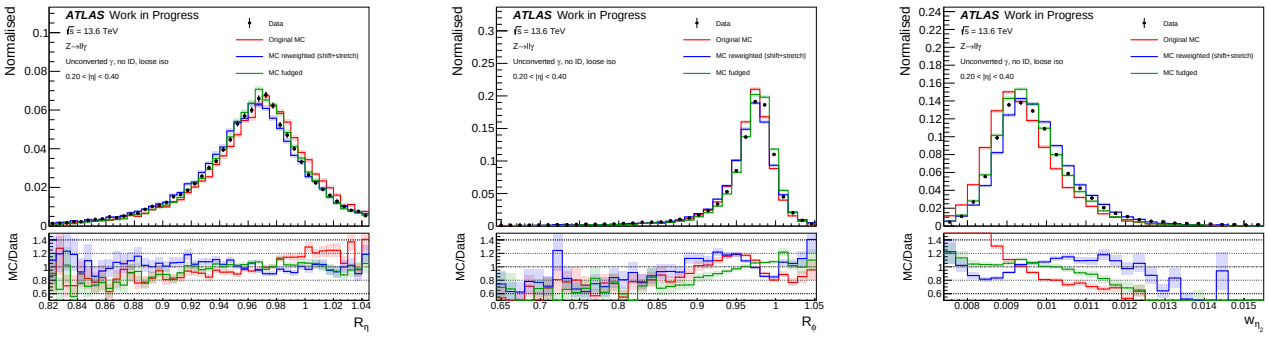
$|\eta| \in [0.00, 0.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 16: Shower shapes vs. fudge factors, $|\eta| \in [0.00, 0.20)$.

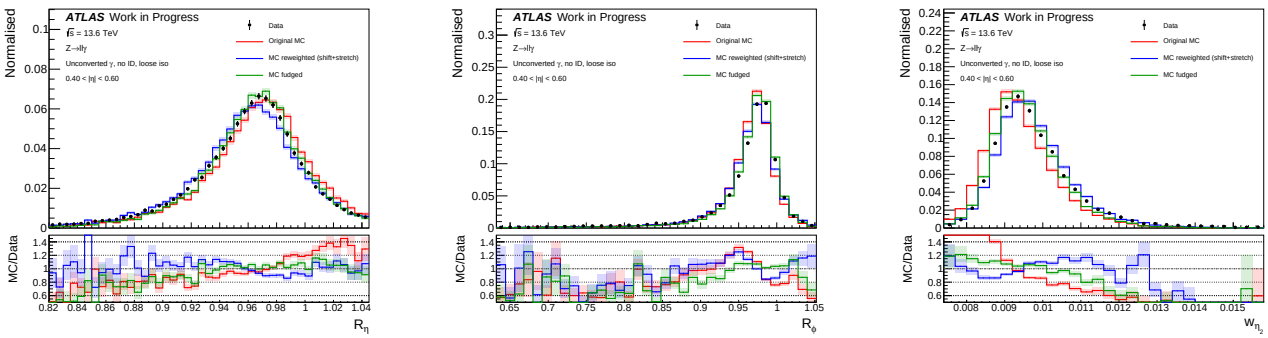
$|\eta| \in [0.20, 0.40)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 17: Shower shapes vs. fudge factors, $|\eta| \in [0.20, 0.40)$.

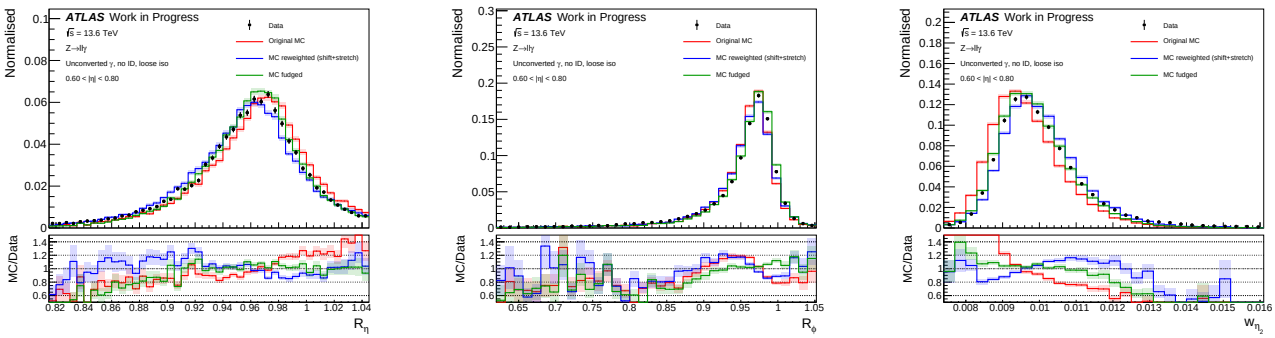
$|\eta| \in [0.40, 0.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 18: Shower shapes vs. fudge factors, $|\eta| \in [0.40, 0.60)$.

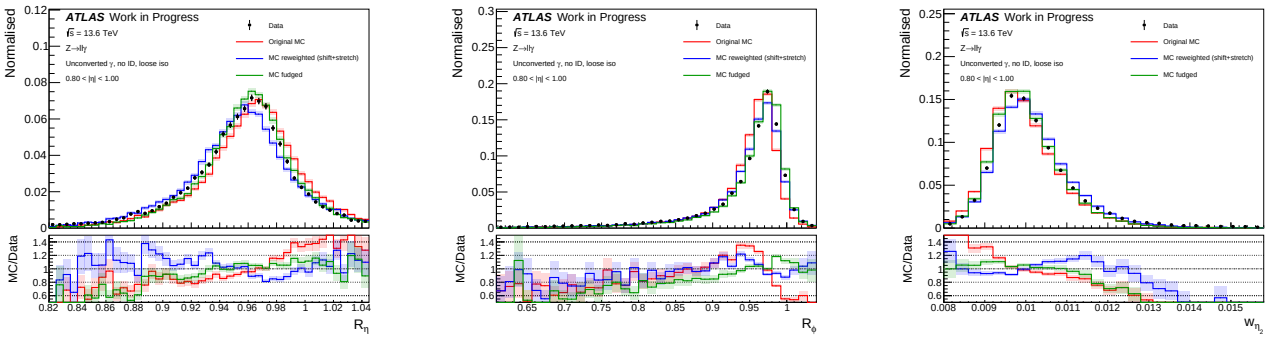
$|\eta| \in [0.60, 0.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 19: Shower shapes vs. fudge factors, $|\eta| \in [0.60, 0.80)$.

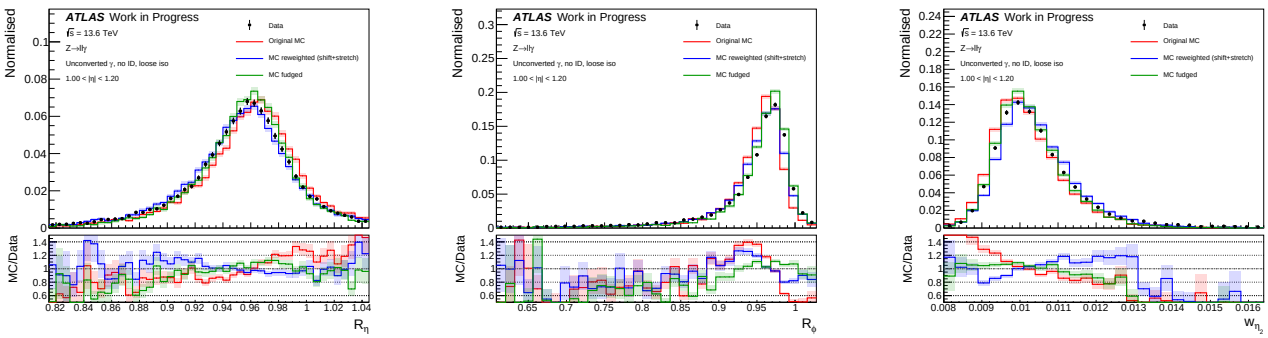
$|\eta| \in [0.80, 1.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 20: Shower shapes vs. fudge factors, $|\eta| \in [0.80, 1.00)$.

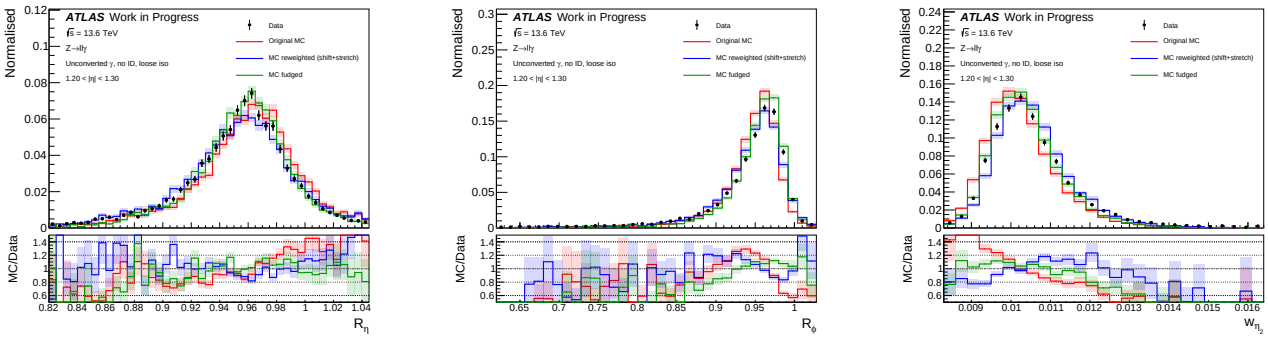
$|\eta| \in [1.00, 1.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 21: Shower shapes vs. fudge factors, $|\eta| \in [1.00, 1.20)$.

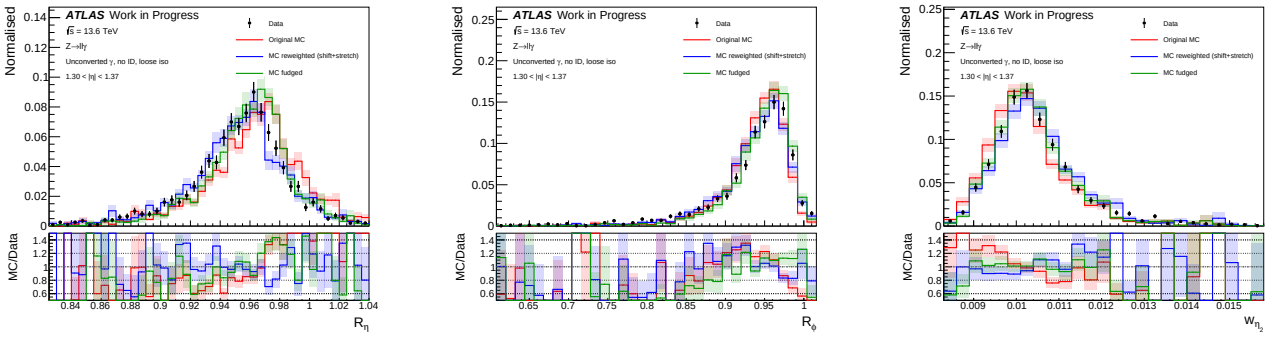
$|\eta| \in [1.20, 1.30)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 22: Shower shapes vs. fudge factors, $|\eta| \in [1.20, 1.30)$.

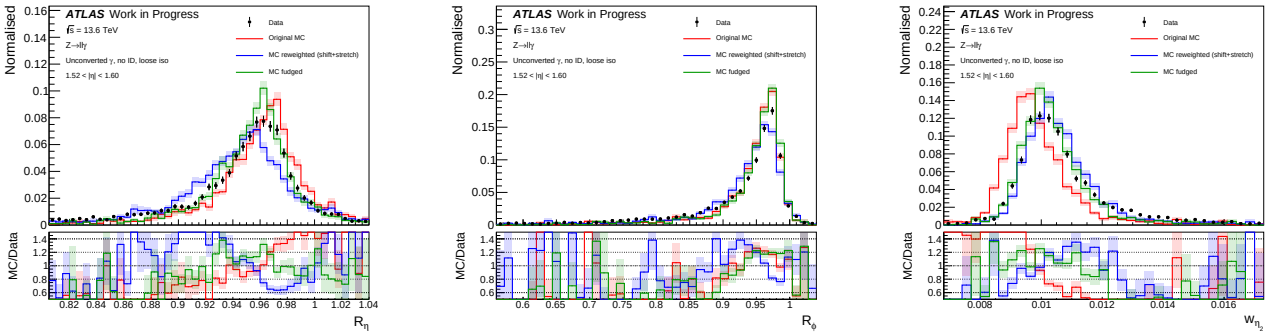
$|\eta| \in [1.30, 1.37)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 23: Shower shapes vs. fudge factors, $|\eta| \in [1.30, 1.37)$.

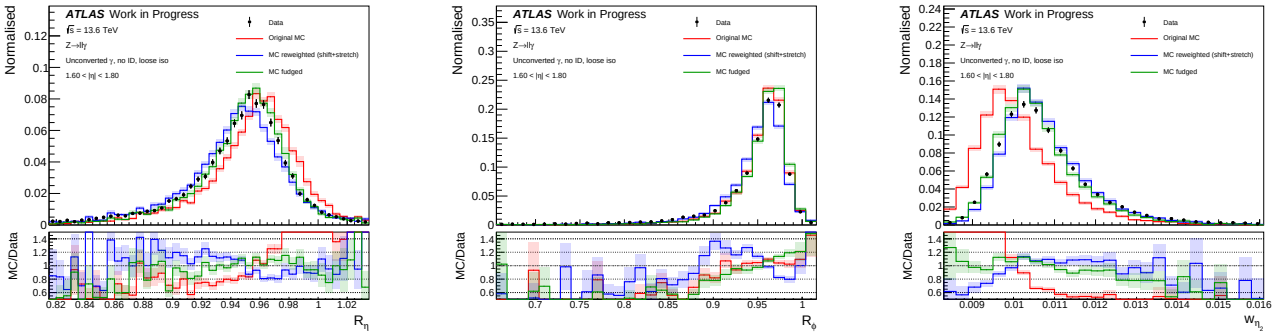
$|\eta| \in [1.52, 1.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 24: Shower shapes vs. fudge factors, $|\eta| \in [1.52, 1.60)$.

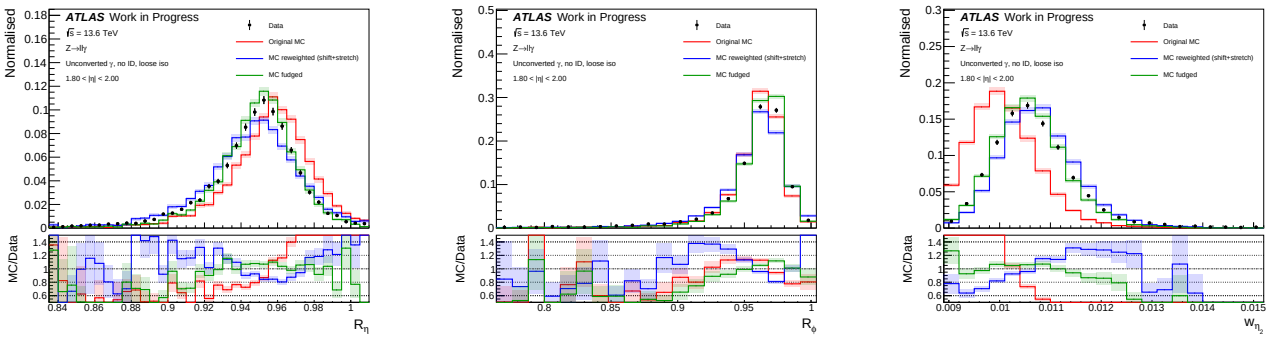
$|\eta| \in [1.60, 1.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 25: Shower shapes vs. fudge factors, $|\eta| \in [1.60, 1.80)$.

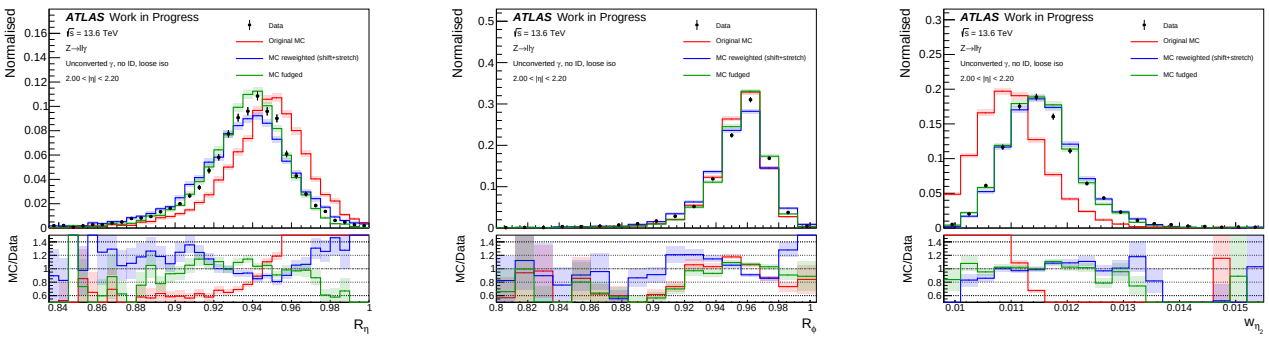
$|\eta| \in [1.80, 2.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 26: Shower shapes vs. fudge factors, $|\eta| \in [1.80, 2.00)$.

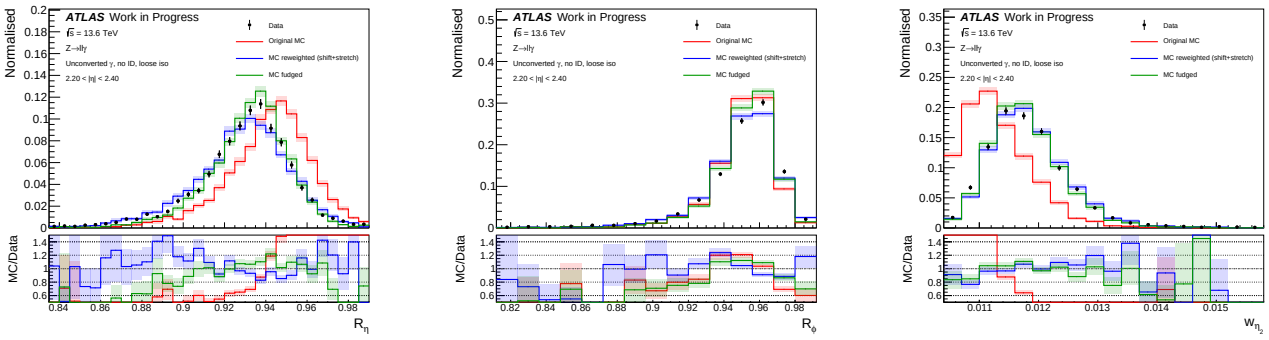
$|\eta| \in [2.00, 2.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 27: Shower shapes vs. fudge factors, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$

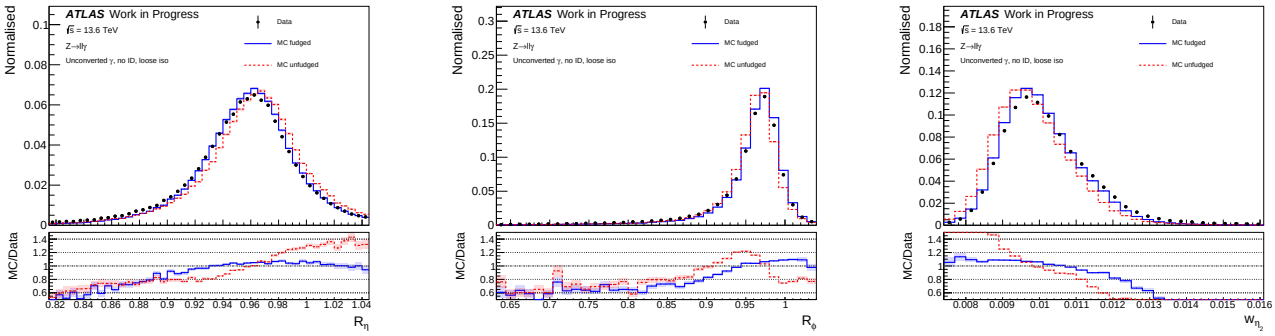


(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 28: Shower shapes vs. fudge factors, $|\eta| \in [2.20, 2.40)$.

5 Fudge Factor Comparison (Integrated)

Stored (fudged) branch values versus unfudged MC and data.



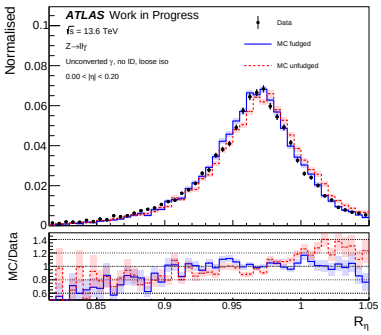
(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 29: Fudge factor comparison: Data (stored) vs MC (fudged) vs MC (unfudged).

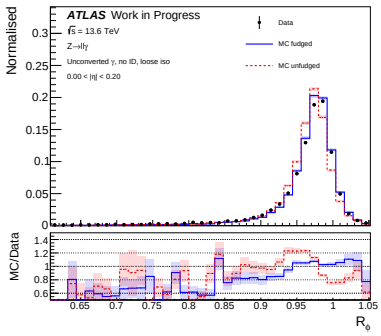
6 Fudge Factor Comparison (Per $|\eta|$ Bin)

Fudge factor comparisons for each $|\eta|$ bin.

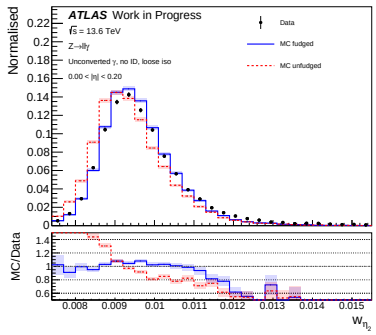
$|\eta| \in [0.00, 0.20)$



(a) R_η



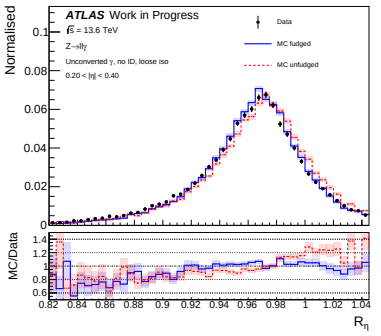
(b) R_ϕ



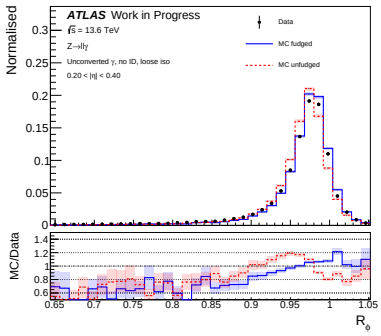
(c) $w_{\eta 2}$

Figure 30: Fudge factor comparison, $|\eta| \in [0.00, 0.20)$.

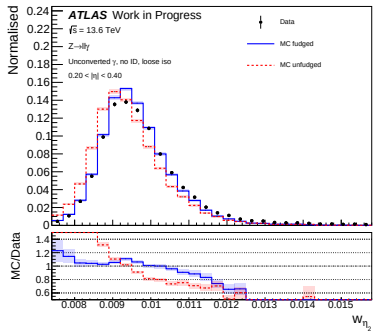
$|\eta| \in [0.20, 0.40)$



(a) R_η



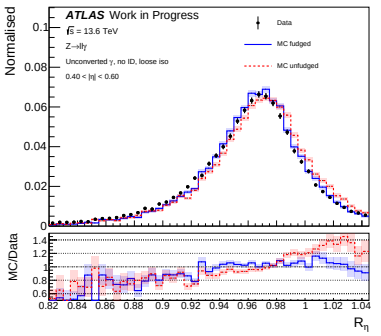
(b) R_ϕ



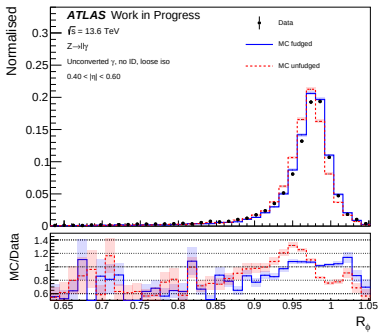
(c) $w_{\eta 2}$

Figure 31: Fudge factor comparison, $|\eta| \in [0.20, 0.40)$.

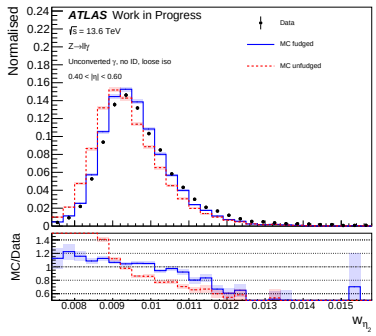
$|\eta| \in [0.40, 0.60)$



(a) R_η



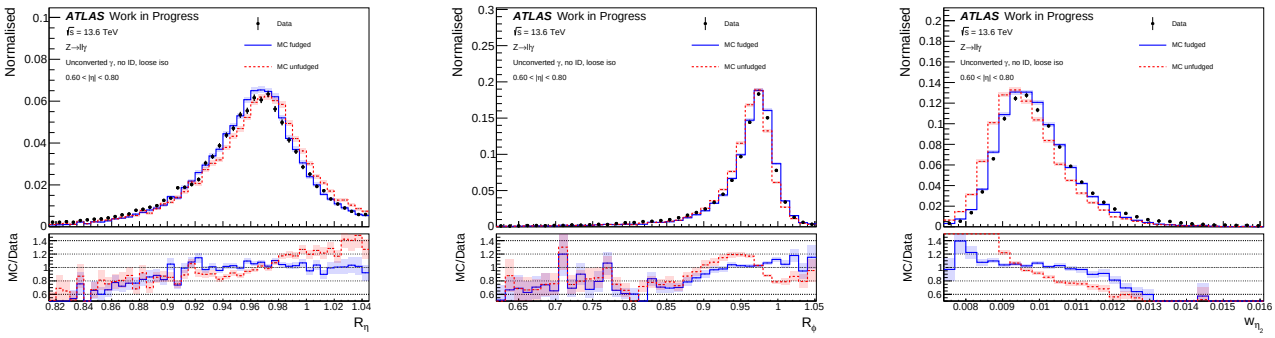
(b) R_ϕ



(c) $w_{\eta 2}$

Figure 32: Fudge factor comparison, $|\eta| \in [0.40, 0.60)$.

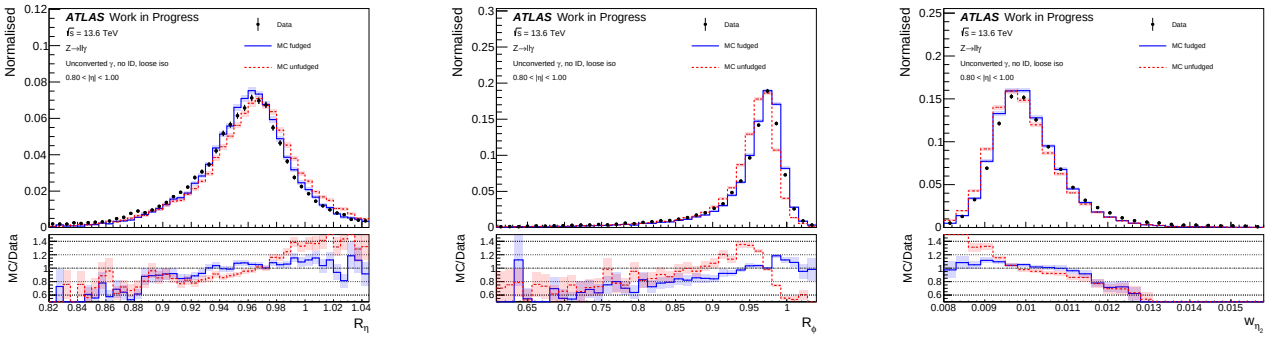
$|\eta| \in [0.60, 0.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 33: Fudge factor comparison, $|\eta| \in [0.60, 0.80)$.

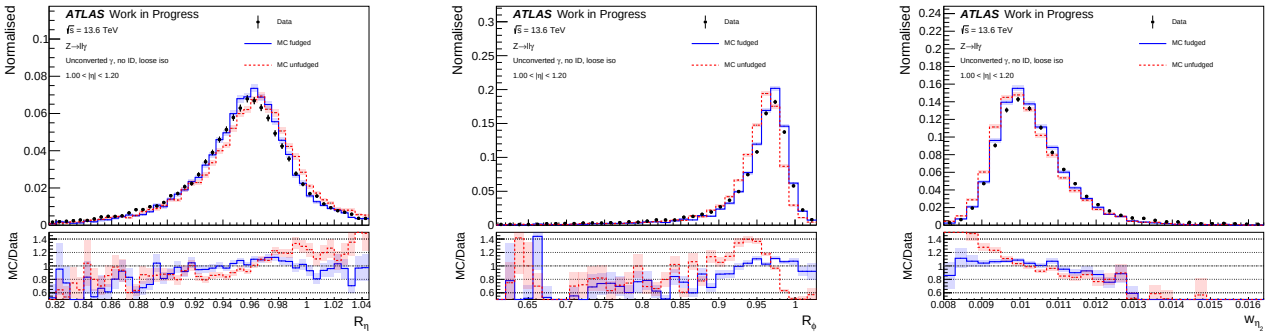
$|\eta| \in [0.80, 1.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 34: Fudge factor comparison, $|\eta| \in [0.80, 1.00)$.

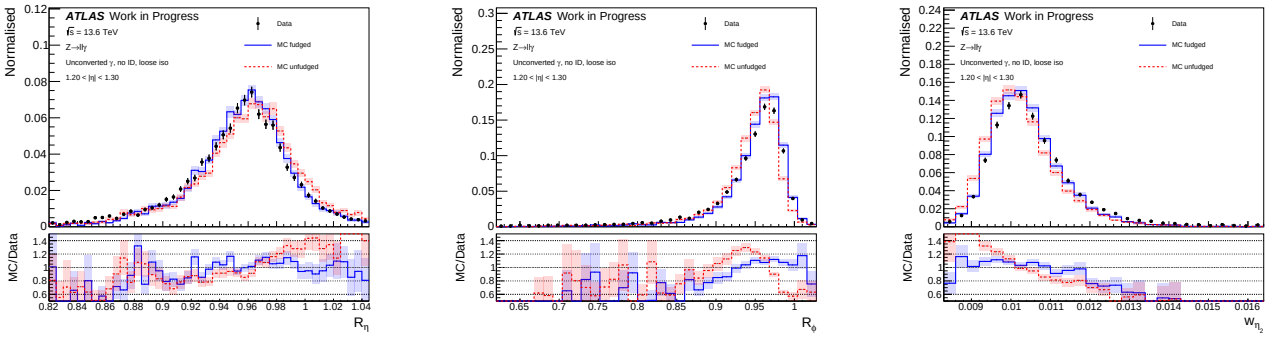
$|\eta| \in [1.00, 1.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 35: Fudge factor comparison, $|\eta| \in [1.00, 1.20)$.

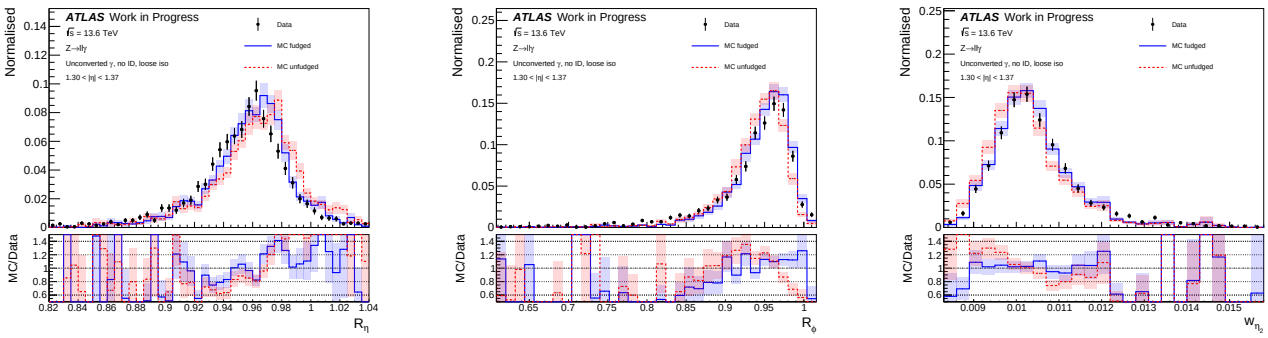
$|\eta| \in [1.20, 1.30)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 36: Fudge factor comparison, $|\eta| \in [1.20, 1.30)$.

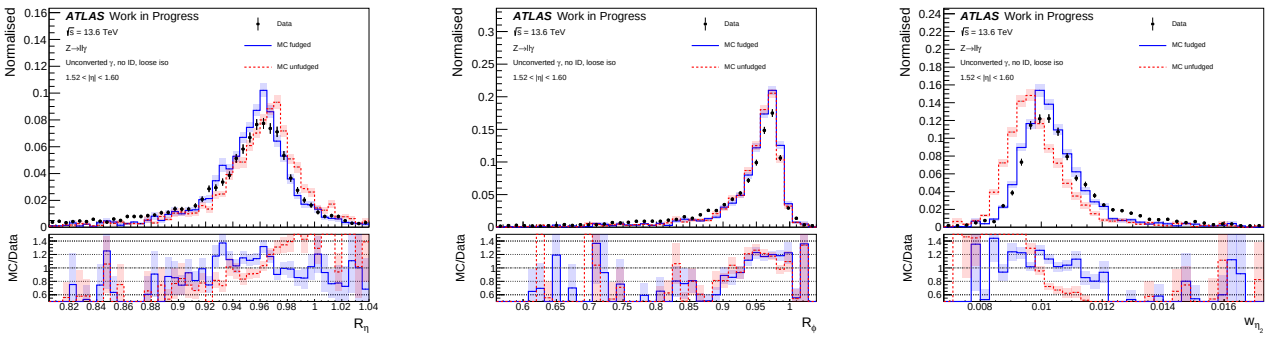
$|\eta| \in [1.30, 1.37)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 37: Fudge factor comparison, $|\eta| \in [1.30, 1.37)$.

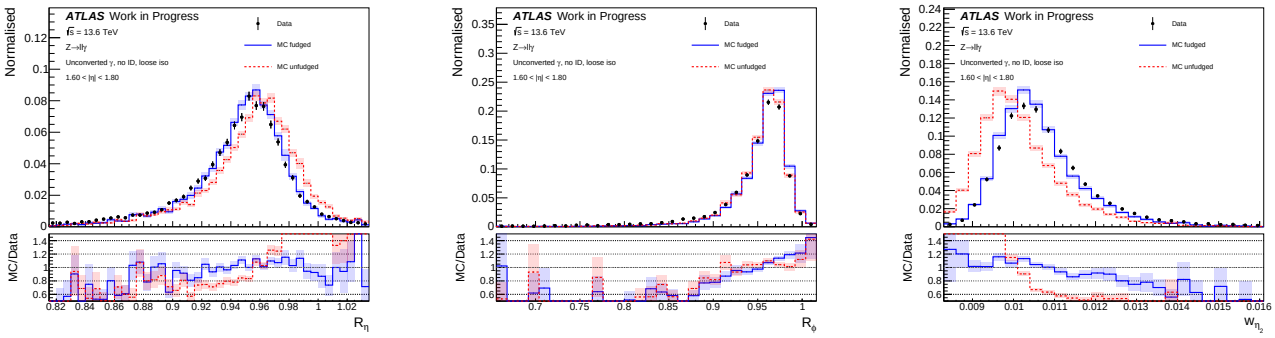
$|\eta| \in [1.52, 1.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 38: Fudge factor comparison, $|\eta| \in [1.52, 1.60)$.

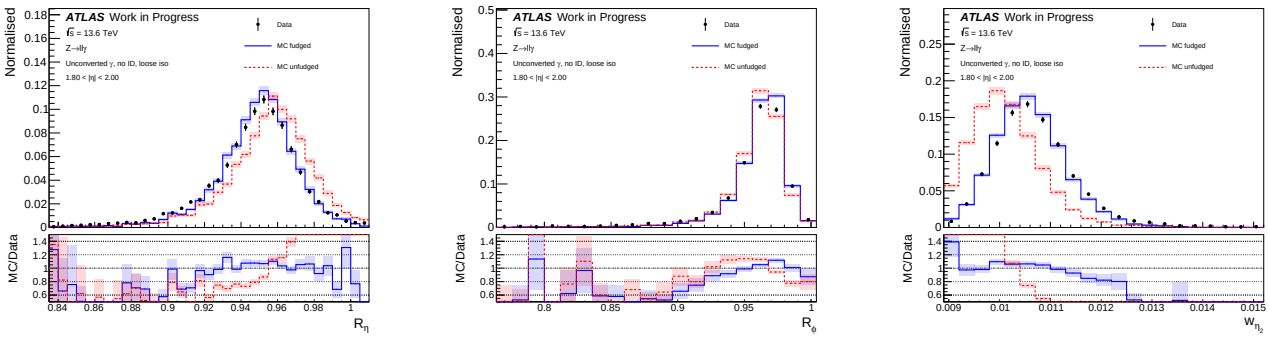
$|\eta| \in [1.60, 1.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 39: Fudge factor comparison, $|\eta| \in [1.60, 1.80)$.

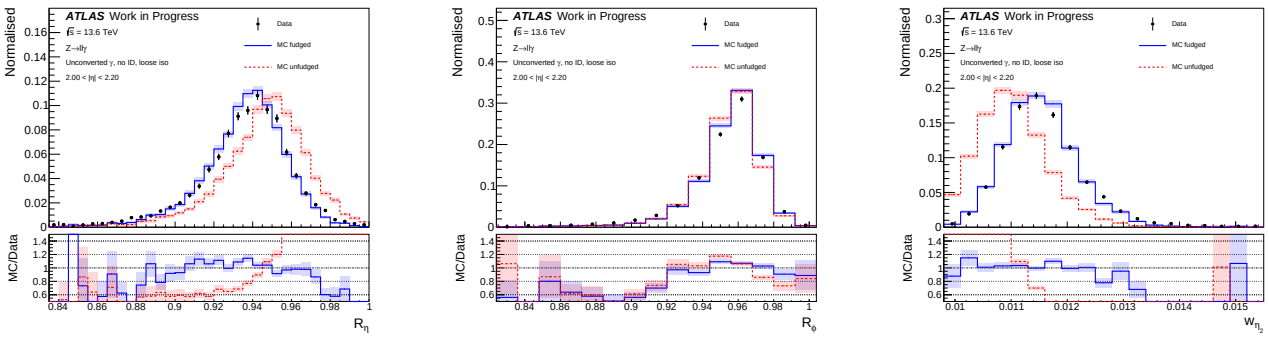
$|\eta| \in [1.80, 2.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 40: Fudge factor comparison, $|\eta| \in [1.80, 2.00)$.

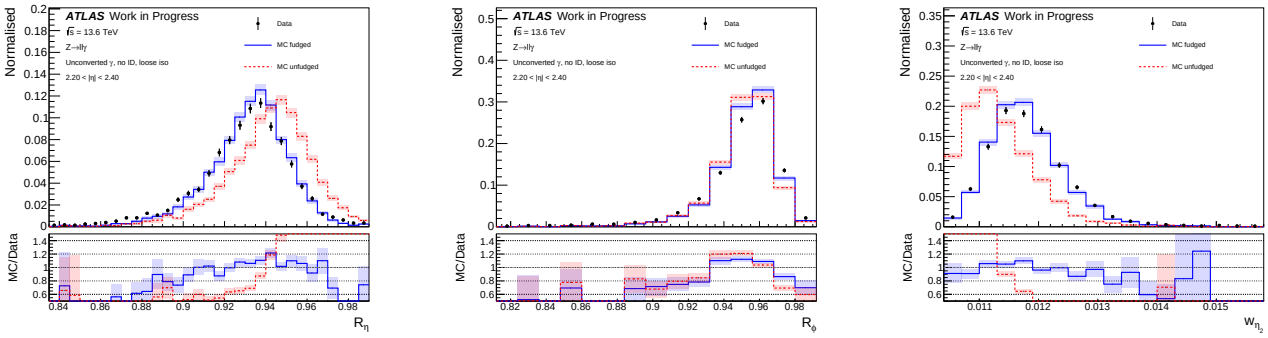
$|\eta| \in [2.00, 2.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 41: Fudge factor comparison, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$

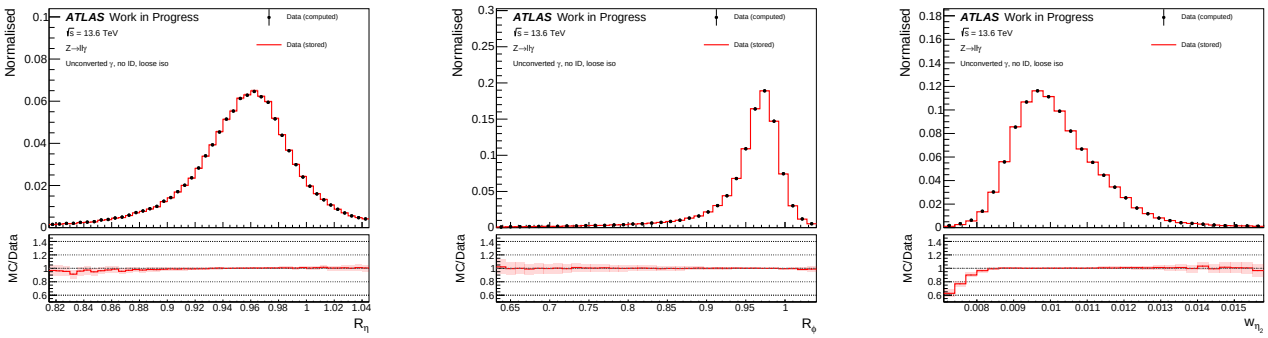


(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 42: Fudge factor comparison, $|\eta| \in [2.20, 2.40)$.

7 Computed vs. Stored Validation (Integrated)

Shower shapes computed from cell energies versus the branch-stored values.



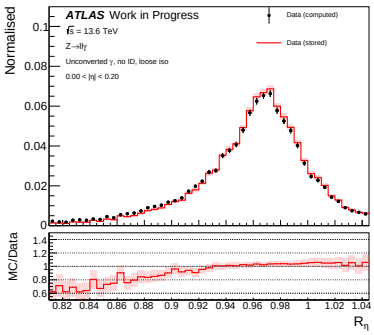
(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 43: Computed (from cells) vs stored (branch) shower shapes.

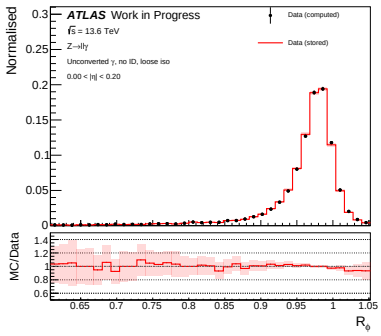
8 Computed vs. Stored (Per $|\eta|$ Bin)

Cell-computed vs branch-stored data per $|\eta|$ bin.

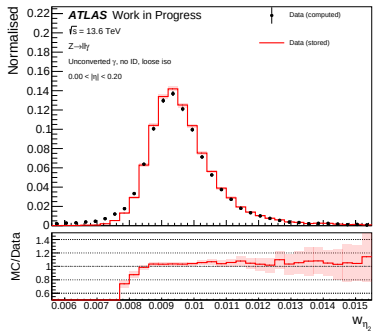
$|\eta| \in [0.00, 0.20)$



(a) R_η



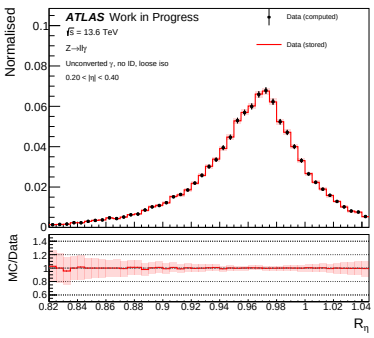
(b) R_ϕ



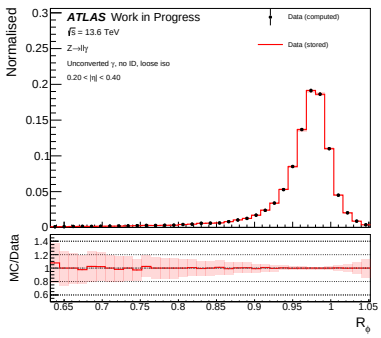
(c) $w_{\eta 2}$

Figure 44: Computed vs. stored, $|\eta| \in [0.00, 0.20)$.

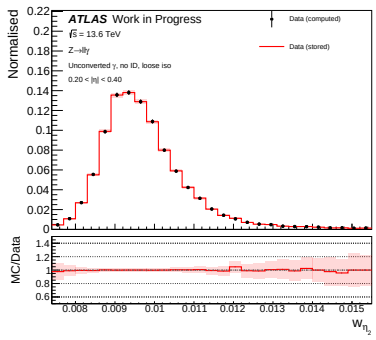
$|\eta| \in [0.20, 0.40)$



(a) R_η



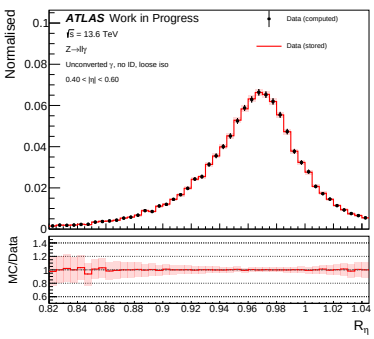
(b) R_ϕ



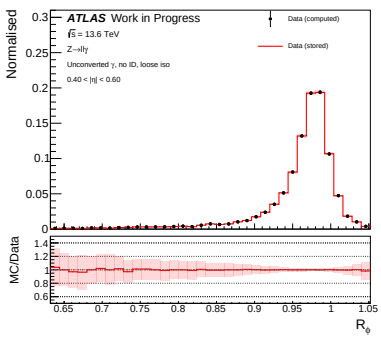
(c) $w_{\eta 2}$

Figure 45: Computed vs. stored, $|\eta| \in [0.20, 0.40)$.

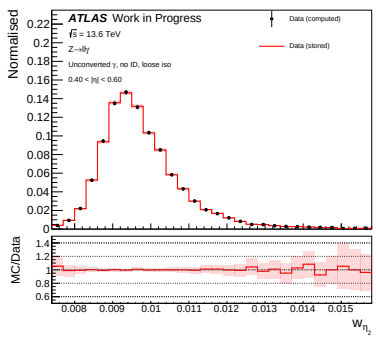
$|\eta| \in [0.40, 0.60)$



(a) R_η



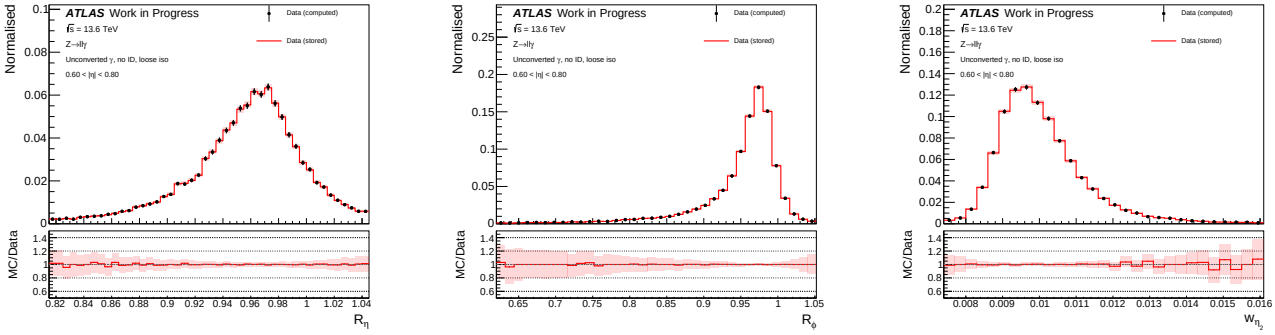
(b) R_ϕ



(c) $w_{\eta 2}$

Figure 46: Computed vs. stored, $|\eta| \in [0.40, 0.60)$.

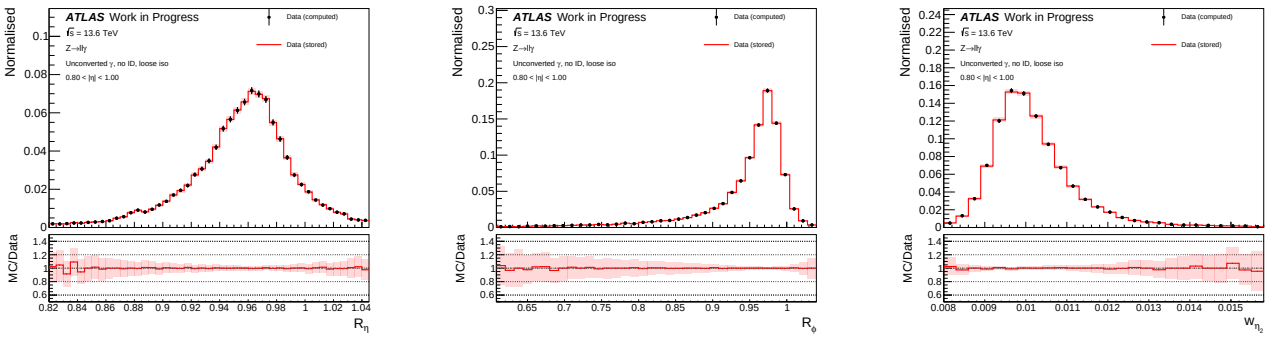
$|\eta| \in [0.60, 0.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 47: Computed vs. stored, $|\eta| \in [0.60, 0.80)$.

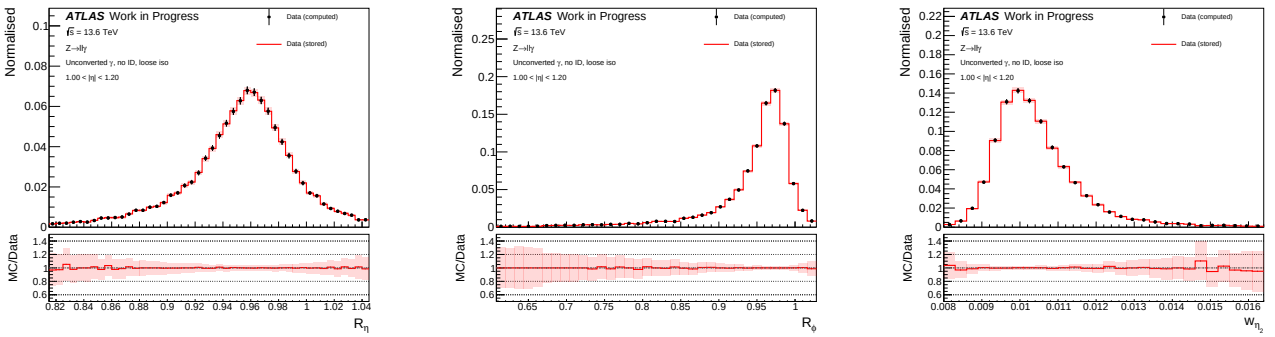
$|\eta| \in [0.80, 1.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 48: Computed vs. stored, $|\eta| \in [0.80, 1.00)$.

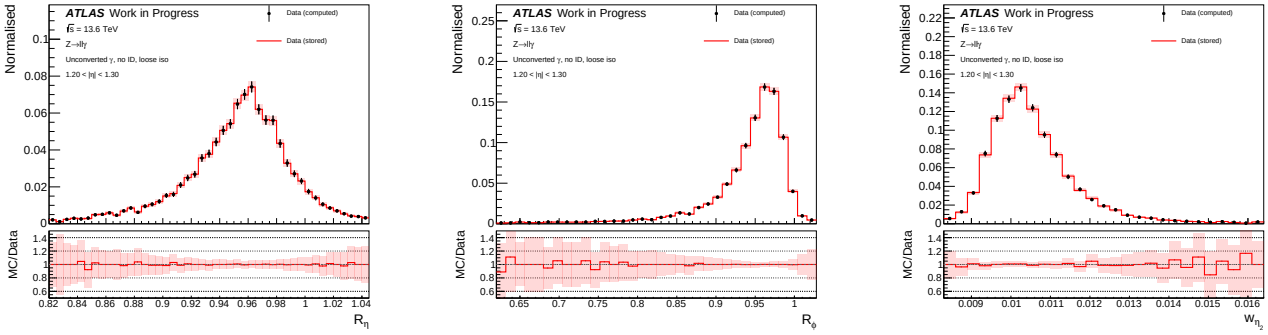
$|\eta| \in [1.00, 1.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 49: Computed vs. stored, $|\eta| \in [1.00, 1.20)$.

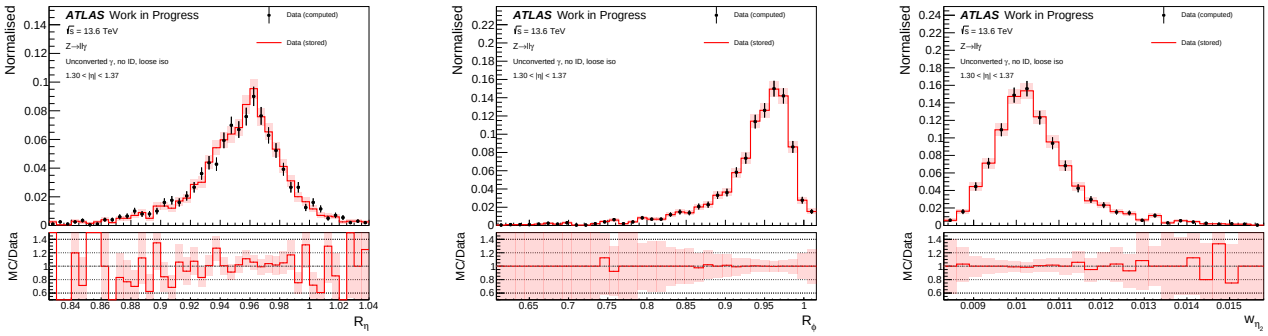
$|\eta| \in [1.20, 1.30)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 50: Computed vs. stored, $|\eta| \in [1.20, 1.30)$.

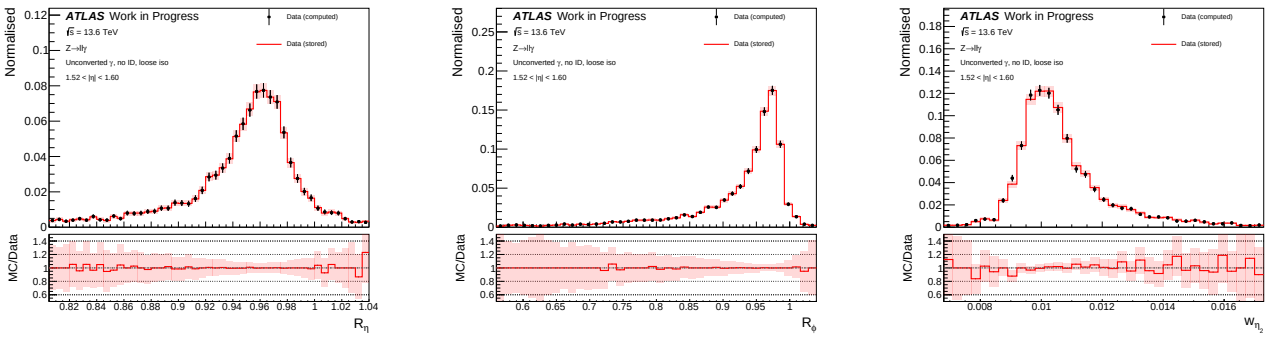
$|\eta| \in [1.30, 1.37)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 51: Computed vs. stored, $|\eta| \in [1.30, 1.37)$.

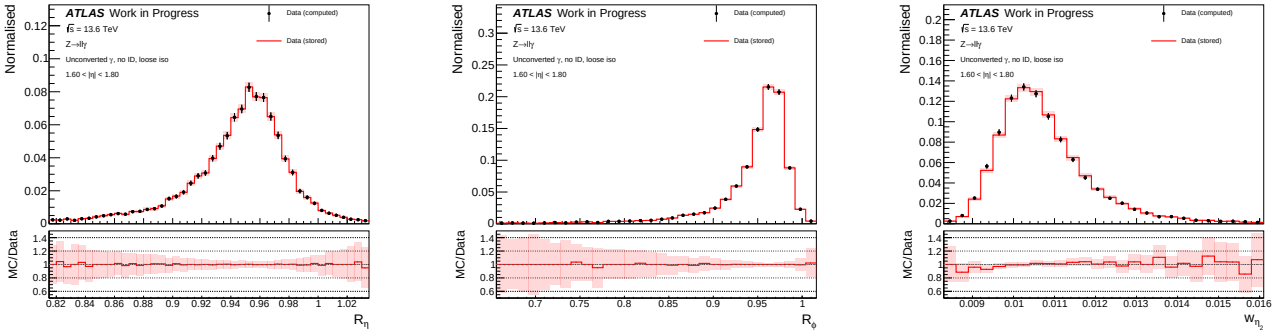
$|\eta| \in [1.52, 1.60)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 52: Computed vs. stored, $|\eta| \in [1.52, 1.60)$.

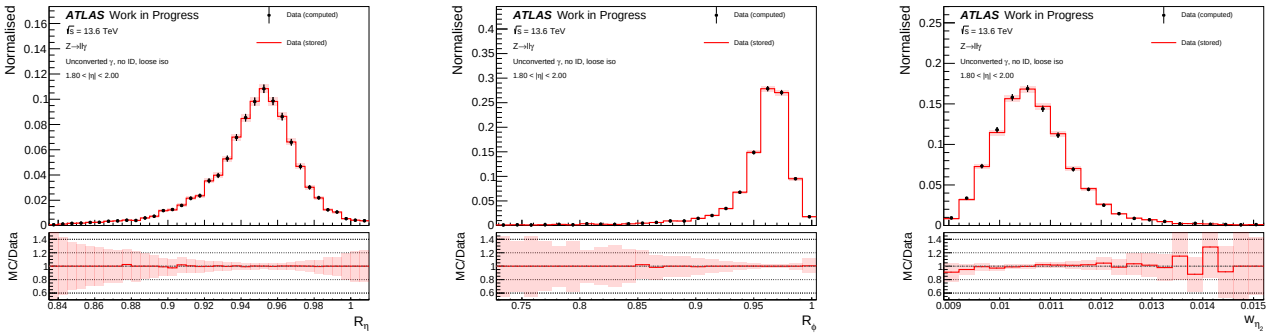
$|\eta| \in [1.60, 1.80)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 53: Computed vs. stored, $|\eta| \in [1.60, 1.80)$.

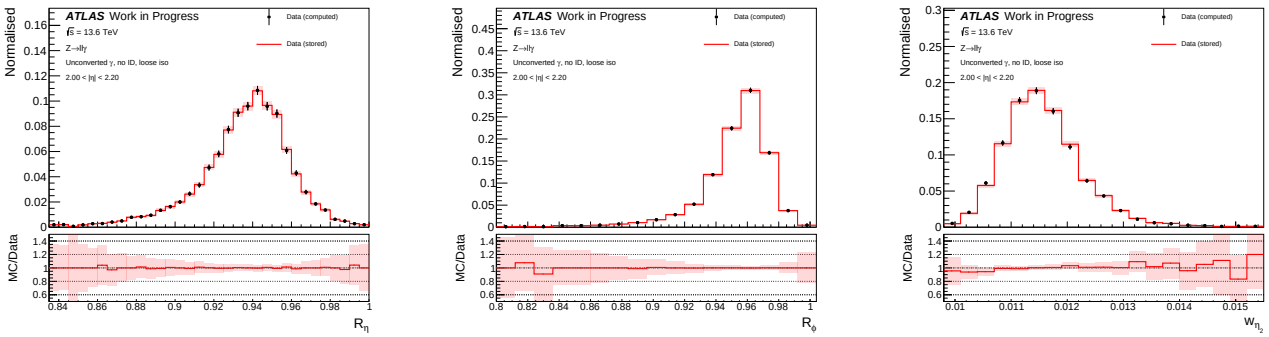
$|\eta| \in [1.80, 2.00)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 54: Computed vs. stored, $|\eta| \in [1.80, 2.00)$.

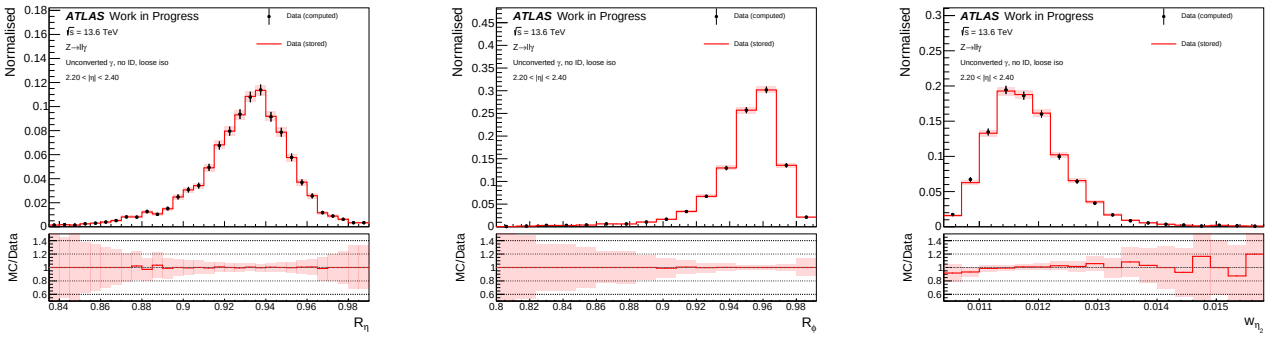
$|\eta| \in [2.00, 2.20)$



(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 55: Computed vs. stored, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$



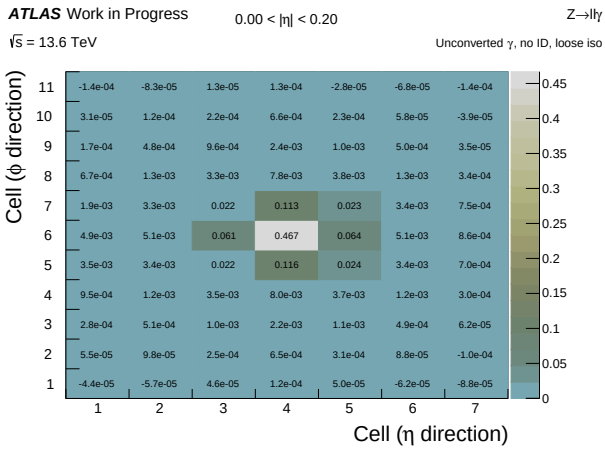
(a) R_η (b) R_ϕ (c) $w_{\eta 2}$

Figure 56: Computed vs. stored, $|\eta| \in [2.20, 2.40)$.

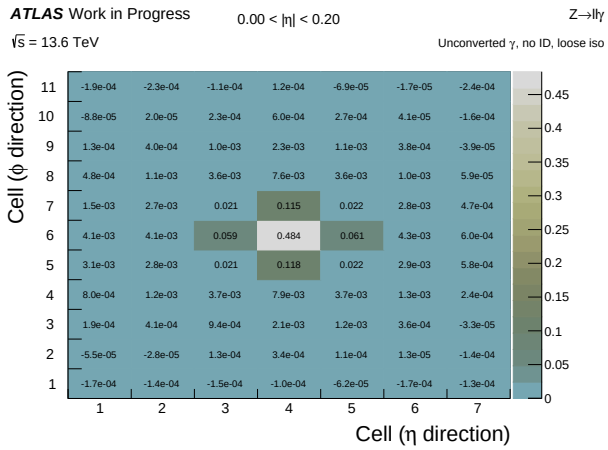
9 Cell Energy Fraction Profiles

Mean cell energy fraction $\langle f_k \rangle$ maps in the 7×11 $(\eta \times \phi)$ grid. Each page shows one $|\eta|$ bin.

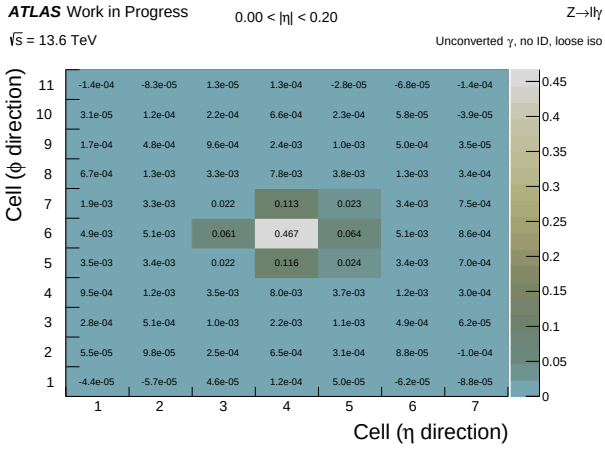
$|\eta| \in [0.00, 0.20)$



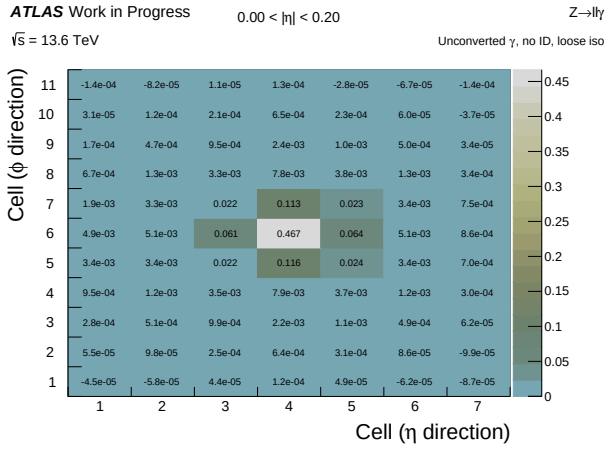
(a) Data



(b) MC (uncorrected)



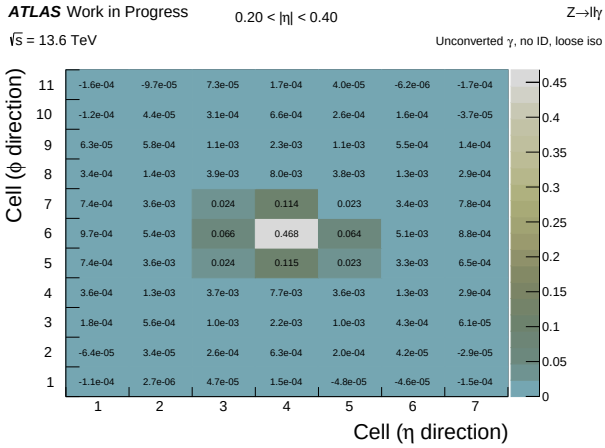
(c) MC after M1



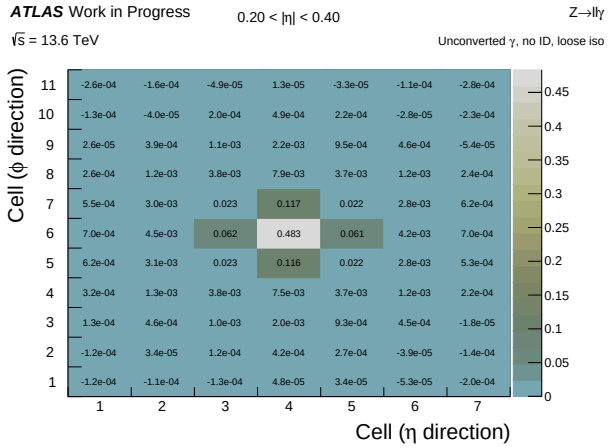
(d) MC after M2

Figure 57: Cell energy fraction maps, $|\eta| \in [0.00, 0.20)$.

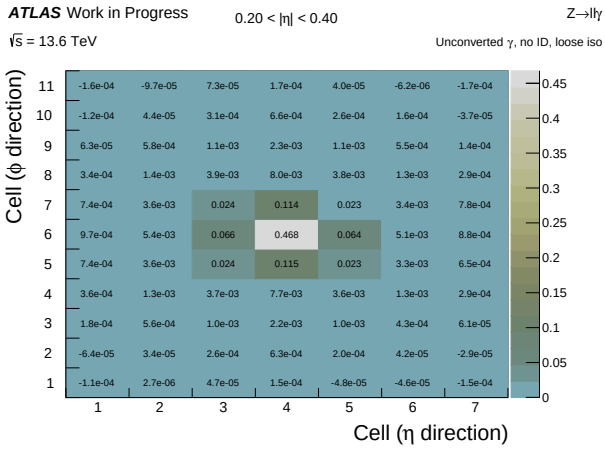
$|\eta| \in [0.20, 0.40)$



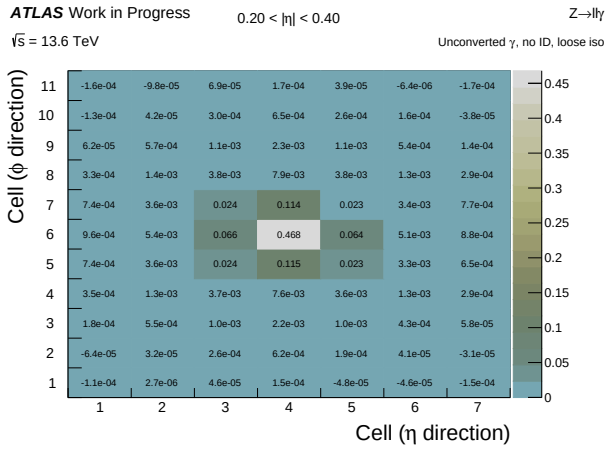
(a) Data



(b) MC (uncorrected)



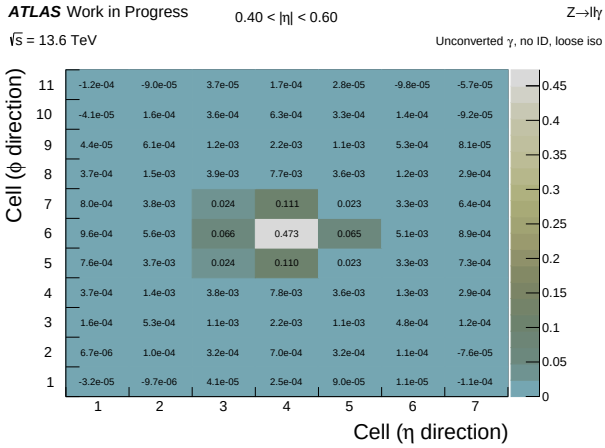
(c) MC after M1



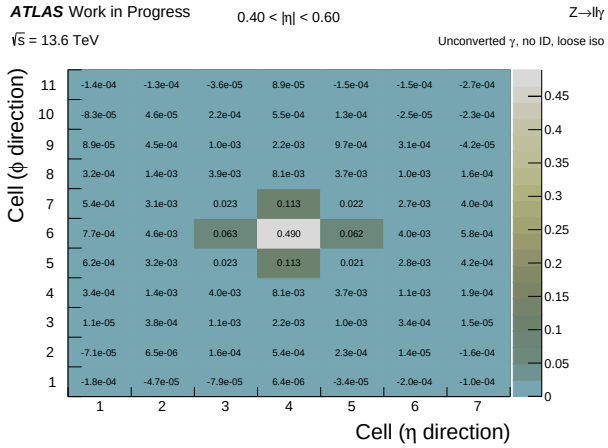
(d) MC after M2

Figure 58: Cell energy fraction maps, $|\eta| \in [0.20, 0.40)$.

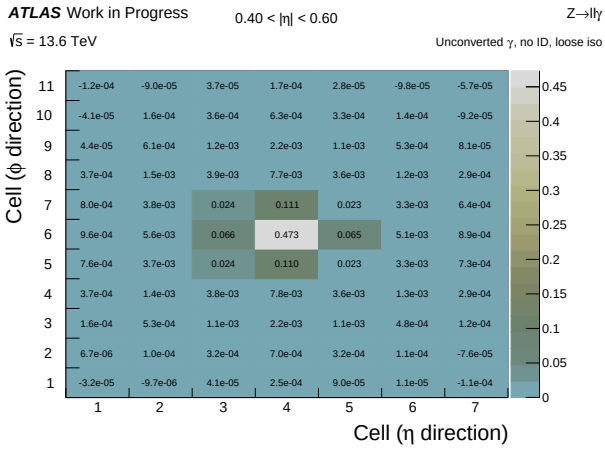
$|\eta| \in [0.40, 0.60)$



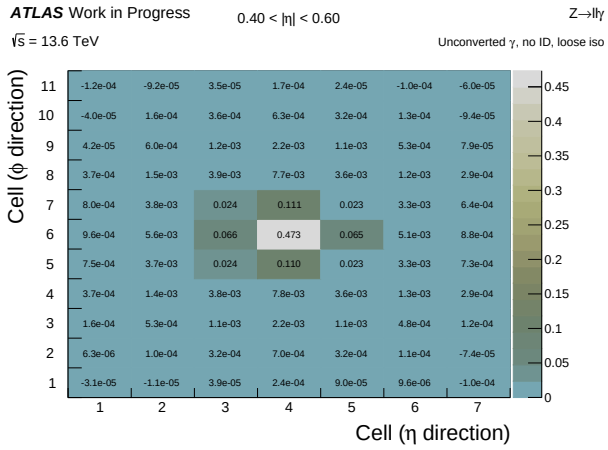
(a) Data



(b) MC (uncorrected)



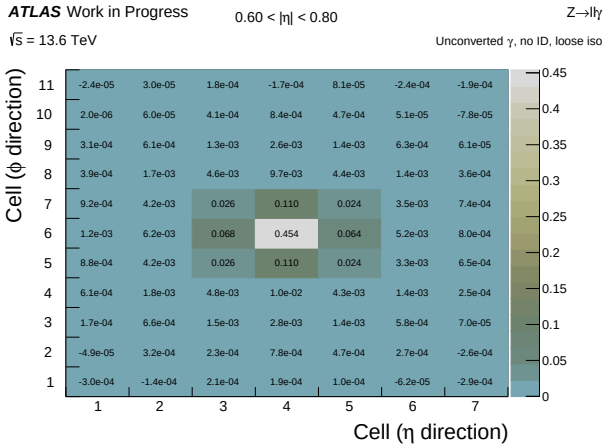
(c) MC after M1



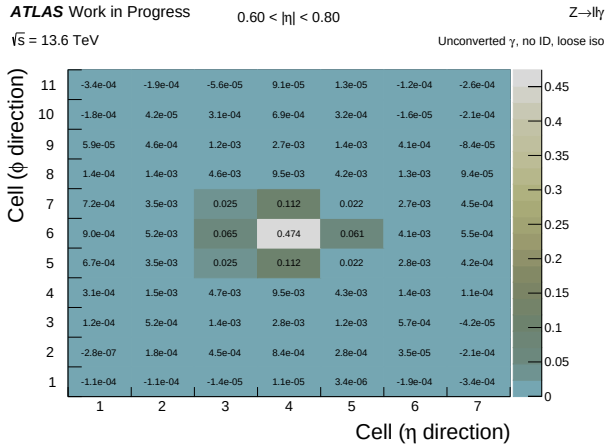
(d) MC after M2

Figure 59: Cell energy fraction maps, $|\eta| \in [0.40, 0.60)$.

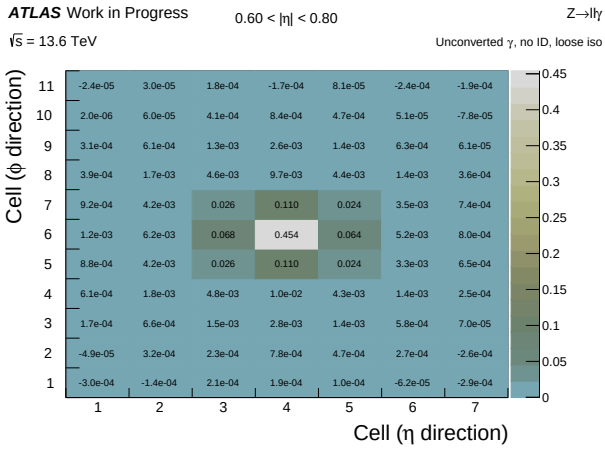
$|\eta| \in [0.60, 0.80)$



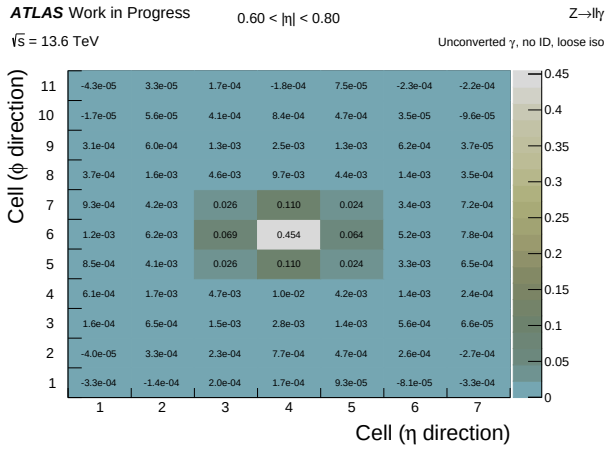
(a) Data



(b) MC (uncorrected)



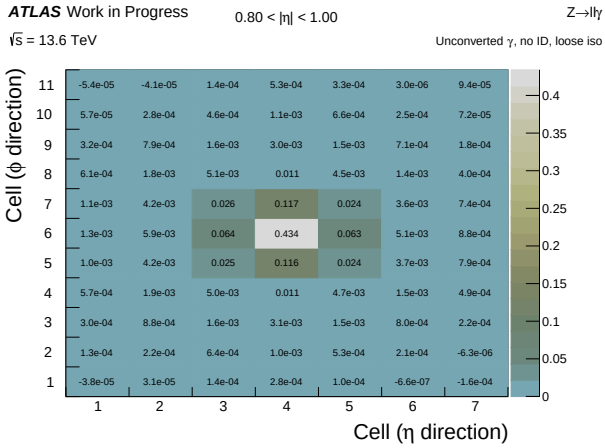
(c) MC after M1



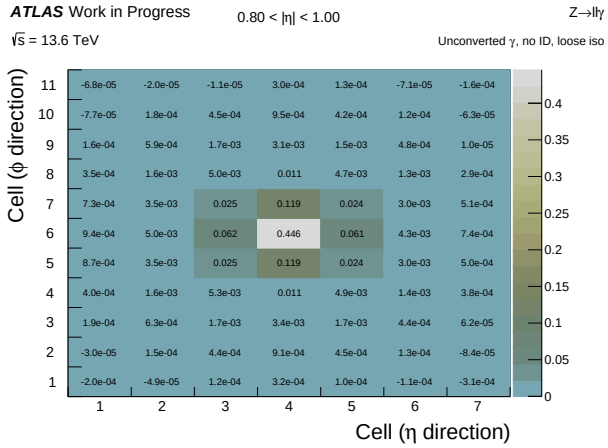
(d) MC after M2

Figure 60: Cell energy fraction maps, $|\eta| \in [0.60, 0.80)$.

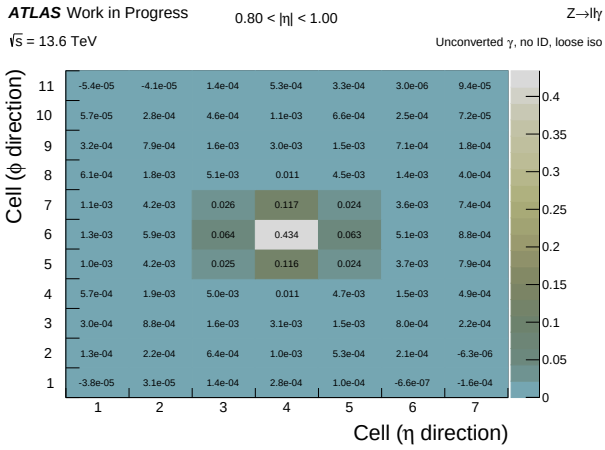
$|\eta| \in [0.80, 1.00)$



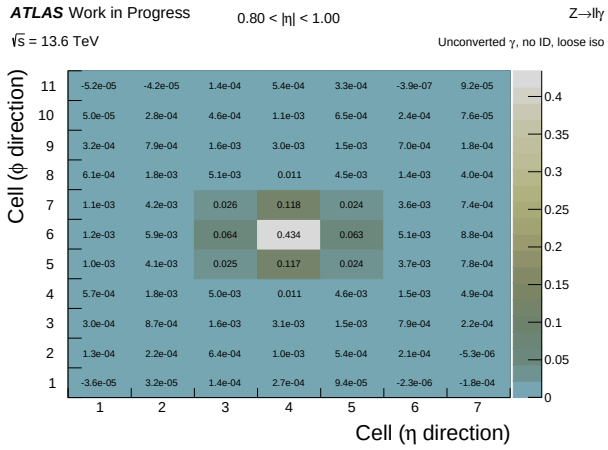
(a) Data



(b) MC (uncorrected)



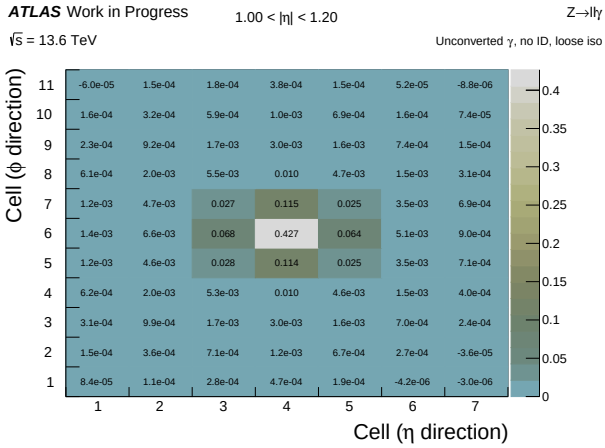
(c) MC after M1



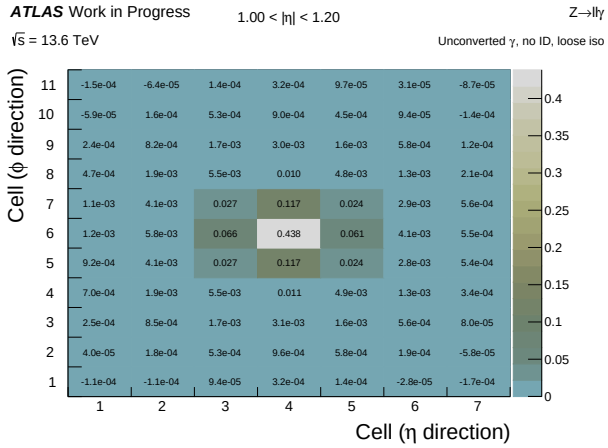
(d) MC after M2

Figure 61: Cell energy fraction maps, $|\eta| \in [0.80, 1.00)$.

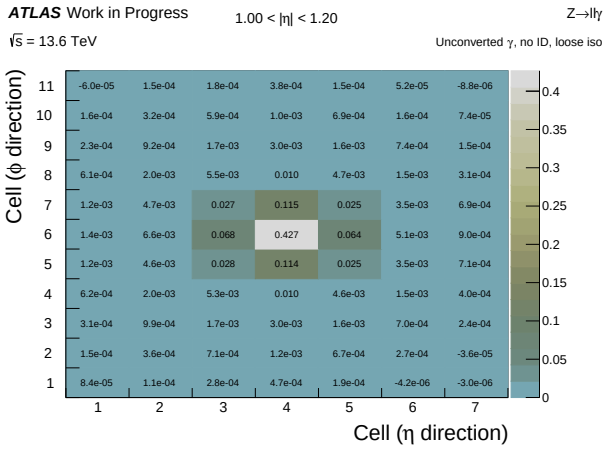
$|\eta| \in [1.00, 1.20)$



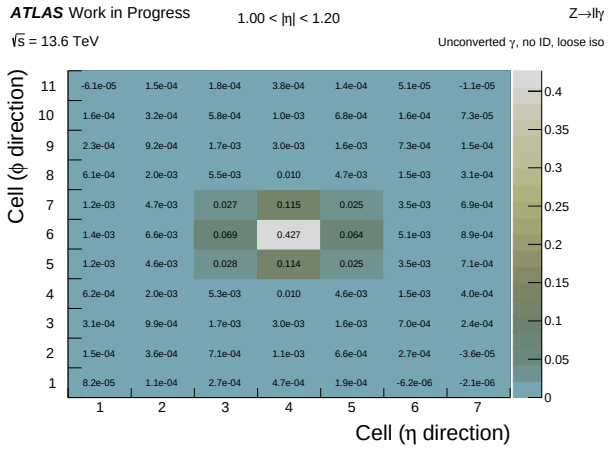
(a) Data



(b) MC (uncorrected)



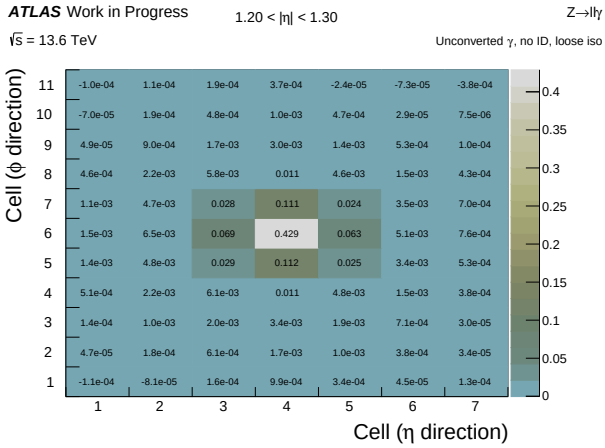
(c) MC after M1



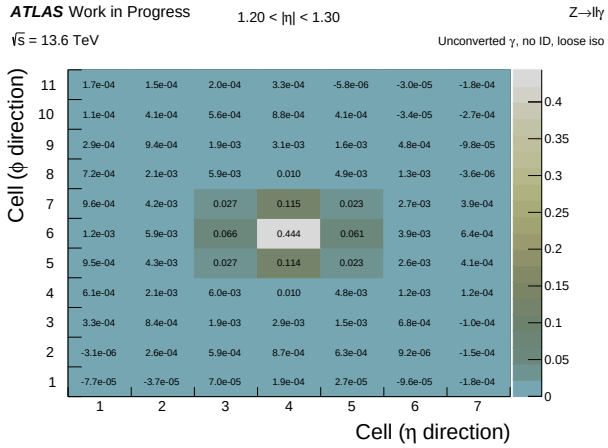
(d) MC after M2

Figure 62: Cell energy fraction maps, $|\eta| \in [1.00, 1.20)$.

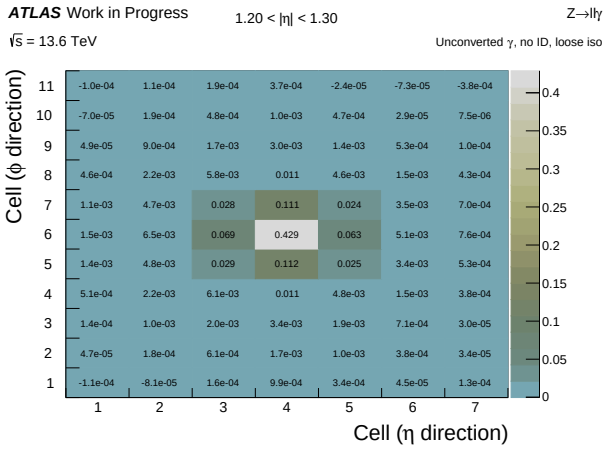
$|\eta| \in [1.20, 1.30)$



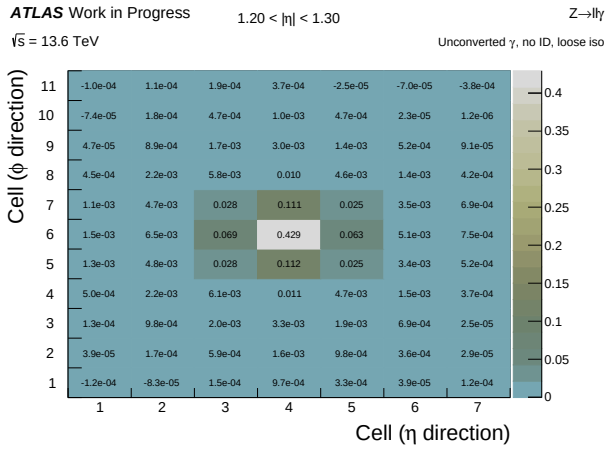
(a) Data



(b) MC (uncorrected)



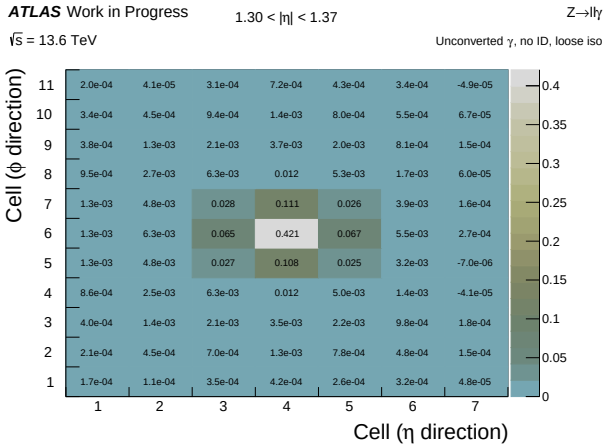
(c) MC after M1



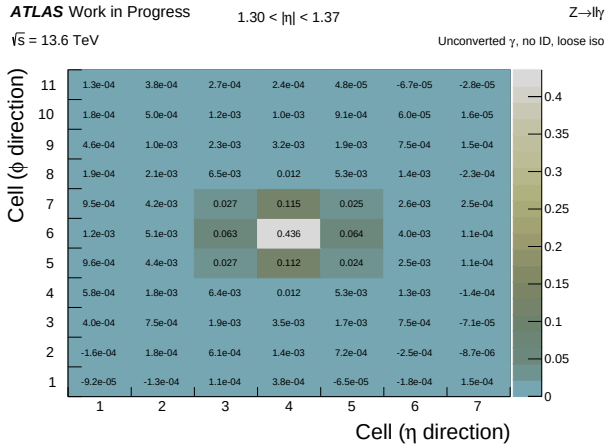
(d) MC after M2

Figure 63: Cell energy fraction maps, $|\eta| \in [1.20, 1.30)$.

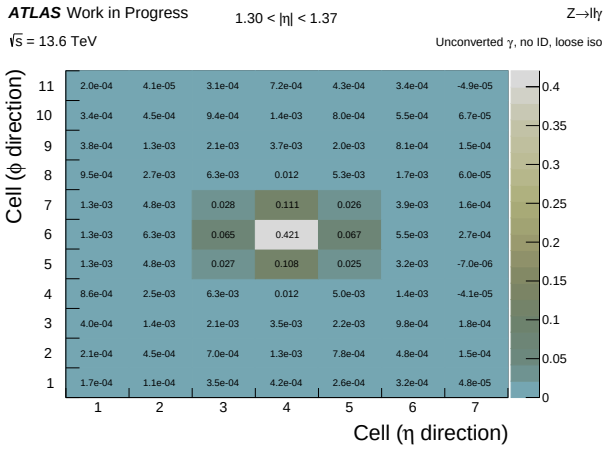
$|\eta| \in [1.30, 1.37)$



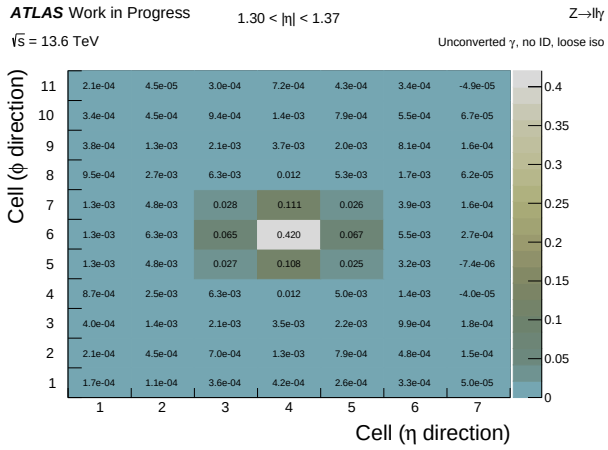
(a) Data



(b) MC (uncorrected)



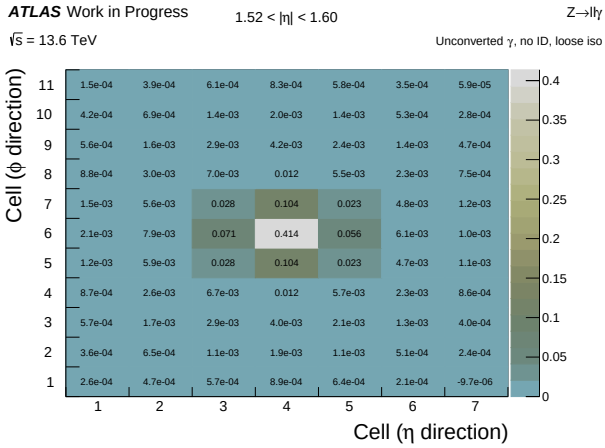
(c) MC after M1



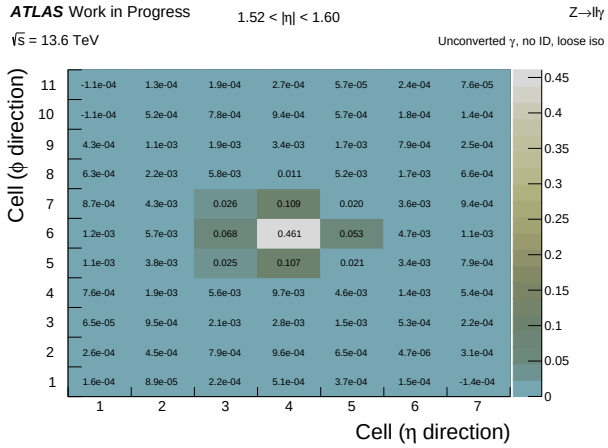
(d) MC after M2

Figure 64: Cell energy fraction maps, $|\eta| \in [1.30, 1.37)$.

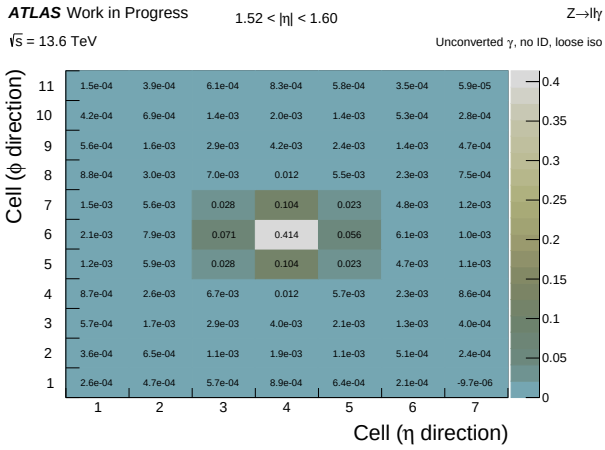
$|\eta| \in [1.52, 1.60)$



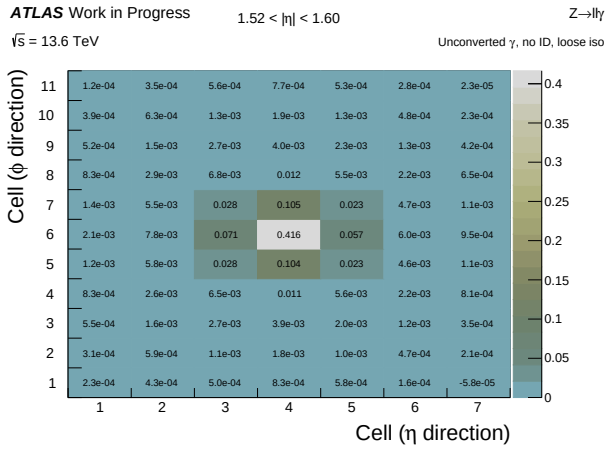
(a) Data



(b) MC (uncorrected)



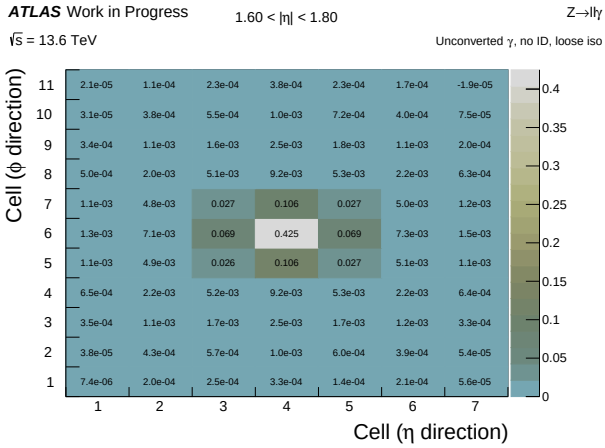
(c) MC after M1



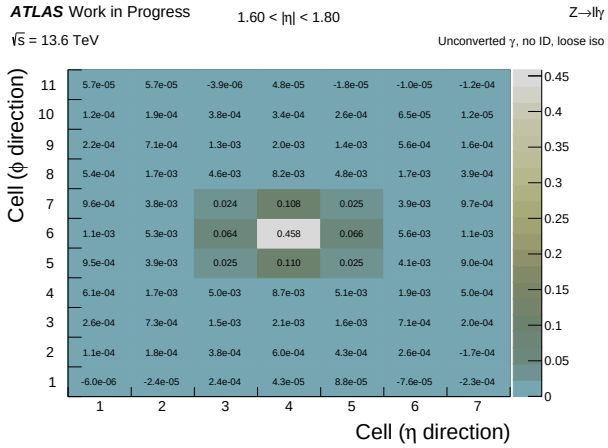
(d) MC after M2

Figure 65: Cell energy fraction maps, $|\eta| \in [1.52, 1.60)$.

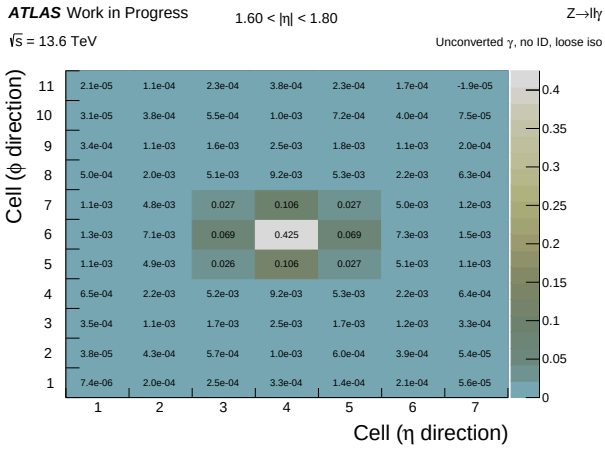
$|\eta| \in [1.60, 1.80)$



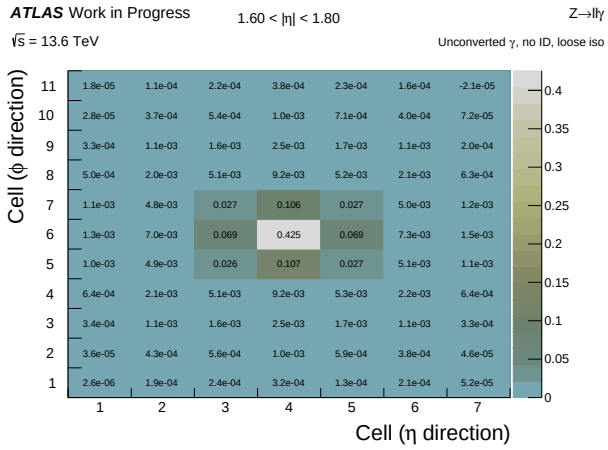
(a) Data



(b) MC (uncorrected)



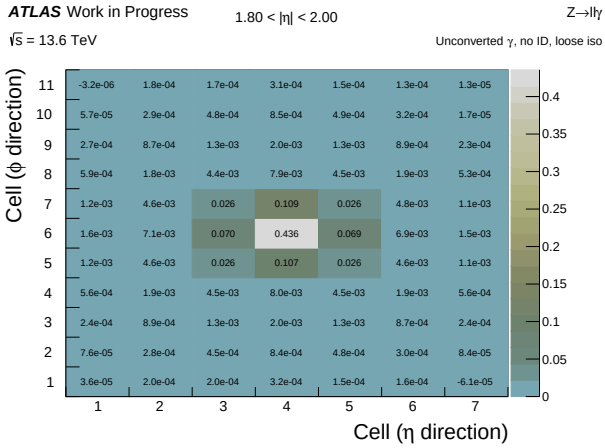
(c) MC after M1



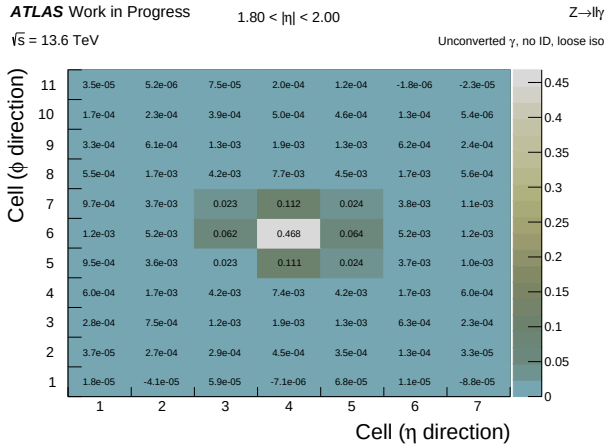
(d) MC after M2

Figure 66: Cell energy fraction maps, $|\eta| \in [1.60, 1.80)$.

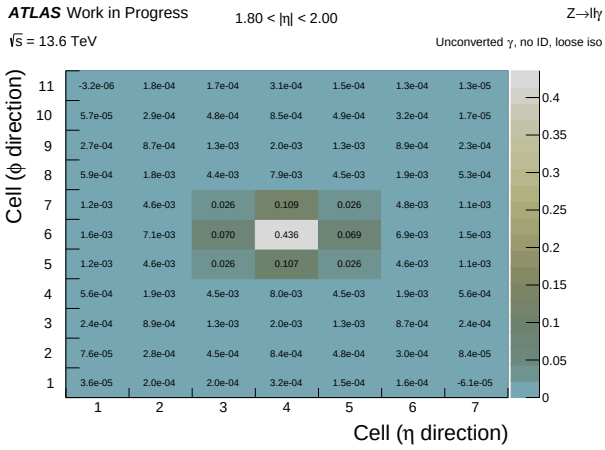
$|\eta| \in [1.80, 2.00)$



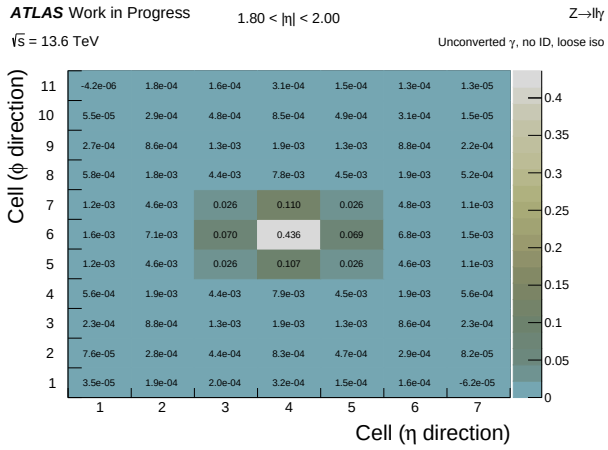
(a) Data



(b) MC (uncorrected)



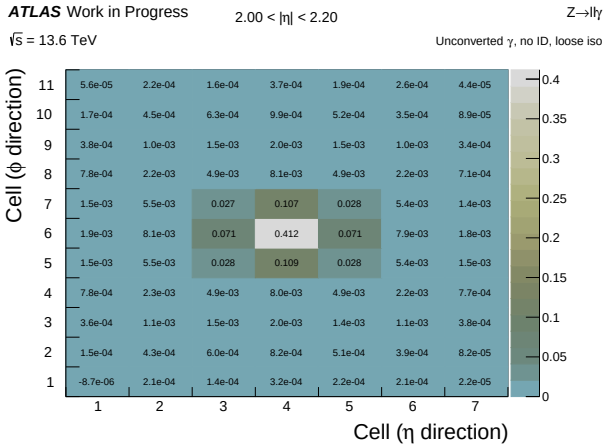
(c) MC after M1



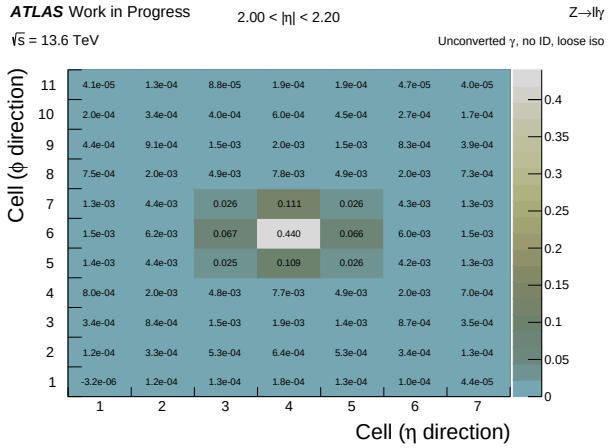
(d) MC after M2

Figure 67: Cell energy fraction maps, $|\eta| \in [1.80, 2.00)$.

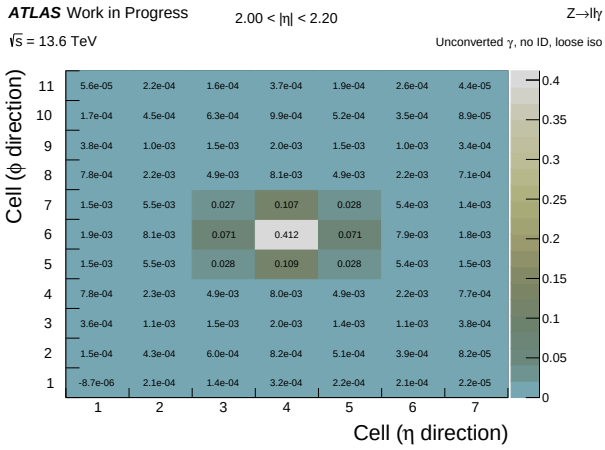
$|\eta| \in [2.00, 2.20)$



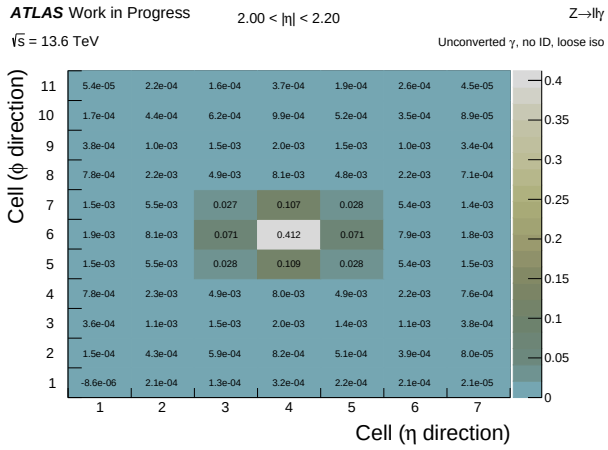
(a) Data



(b) MC (uncorrected)



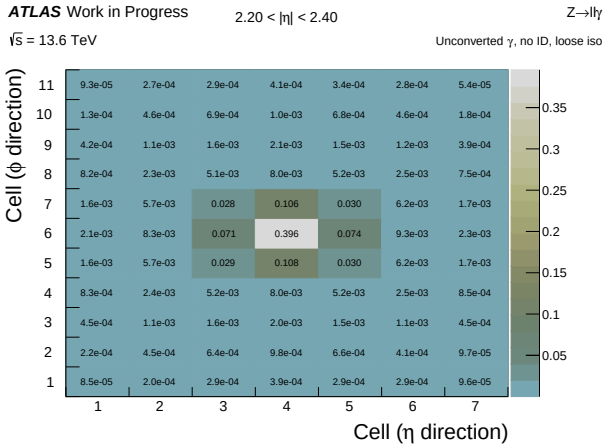
(c) MC after M1



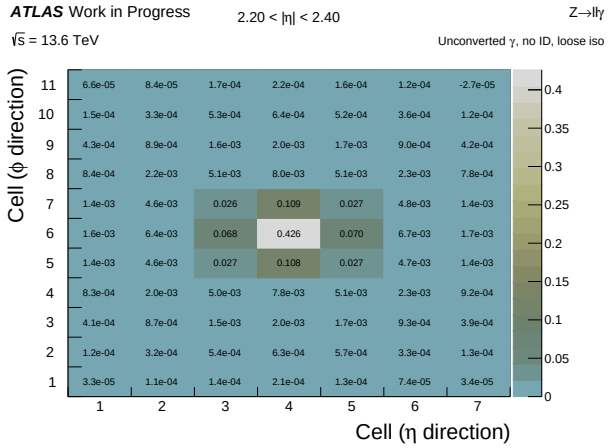
(d) MC after M2

Figure 68: Cell energy fraction maps, $|\eta| \in [2.00, 2.20)$.

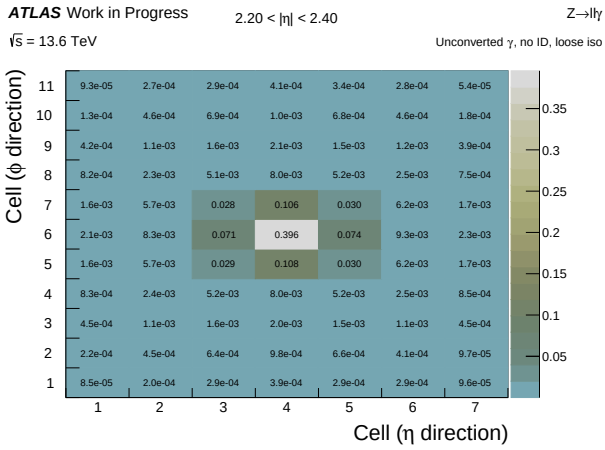
$|\eta| \in [2.20, 2.40)$



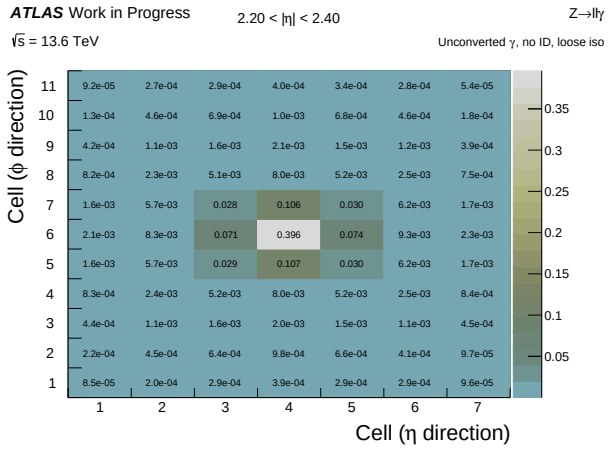
(a) Data



(b) MC (uncorrected)



(c) MC after M1



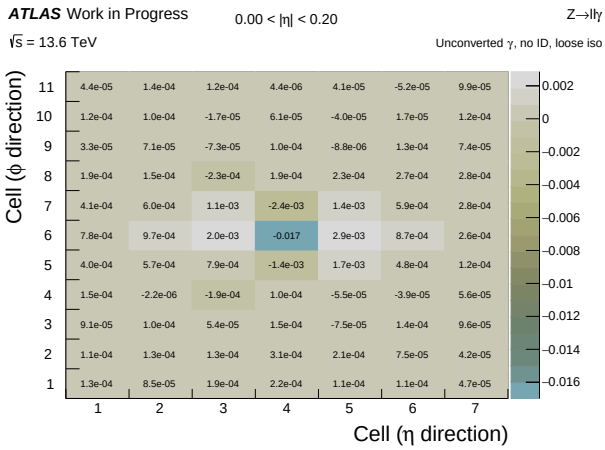
(d) MC after M2

Figure 69: Cell energy fraction maps, $|\eta| \in [2.20, 2.40)$.

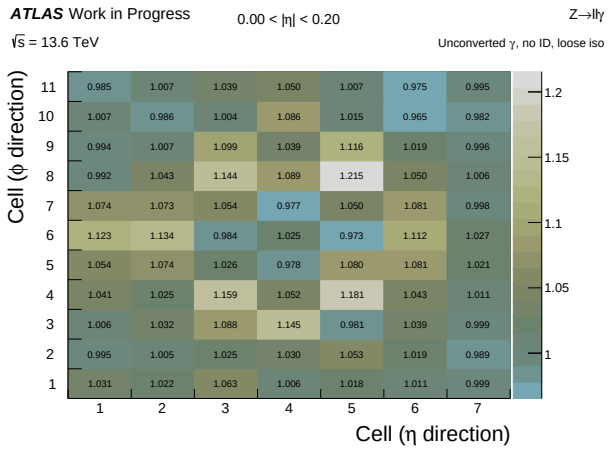
10 Cell-Level Corrections

M1 shift (Δ_k) and M2 stretch (s_k) maps per $|\eta|$ bin.

$|\eta| \in [0.00, 0.20)$



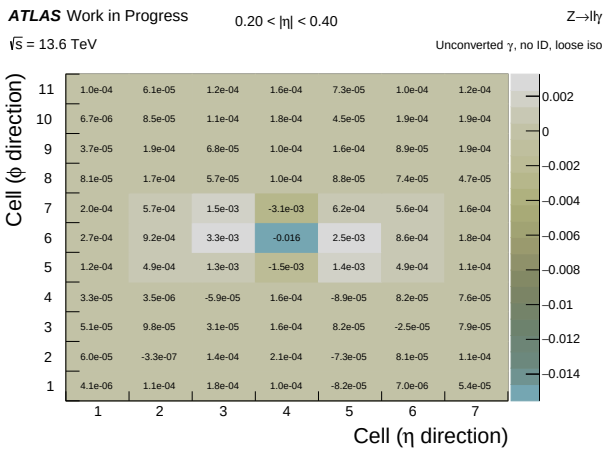
(a) M2 shift (a_k)



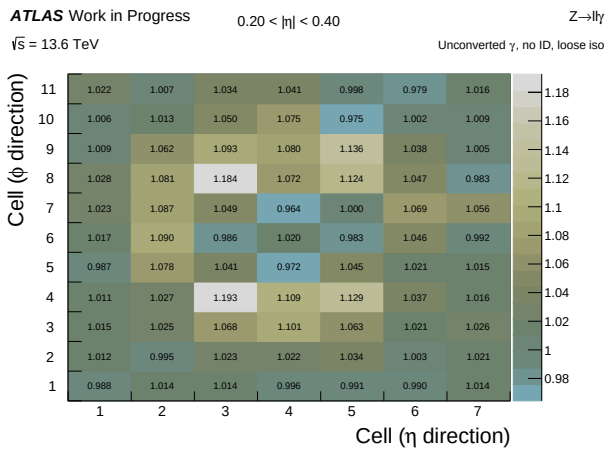
(b) M2 stretch (s_k)

Figure 70: Correction maps, $|\eta| \in [0.00, 0.20)$.

$|\eta| \in [0.20, 0.40)$



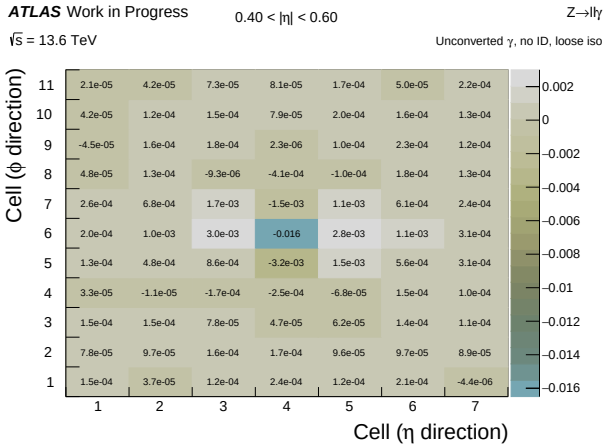
(a) M2 shift (a_k)



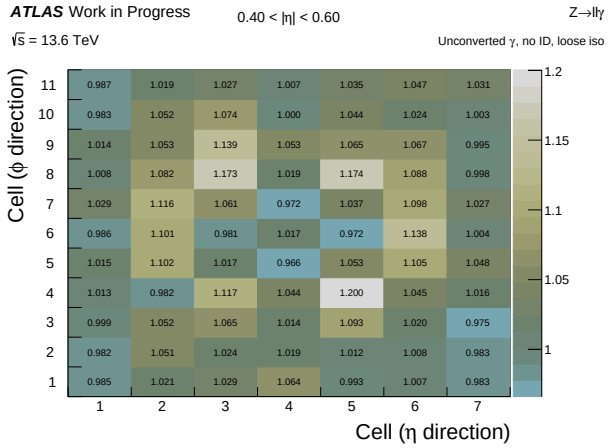
(b) M2 stretch (s_k)

Figure 71: Correction maps, $|\eta| \in [0.20, 0.40)$.

$|\eta| \in [0.40, 0.60)$



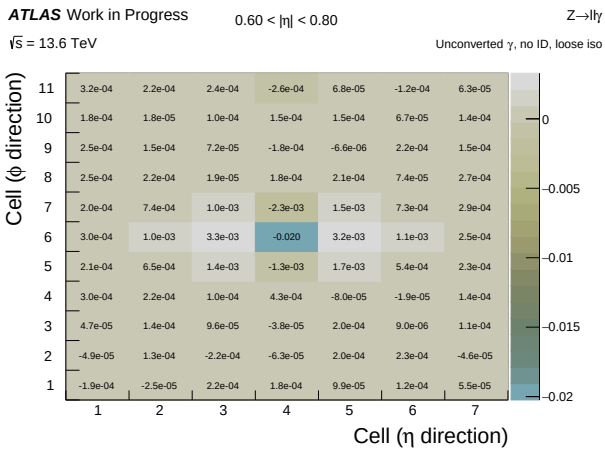
(a) M2 shift (a_k)



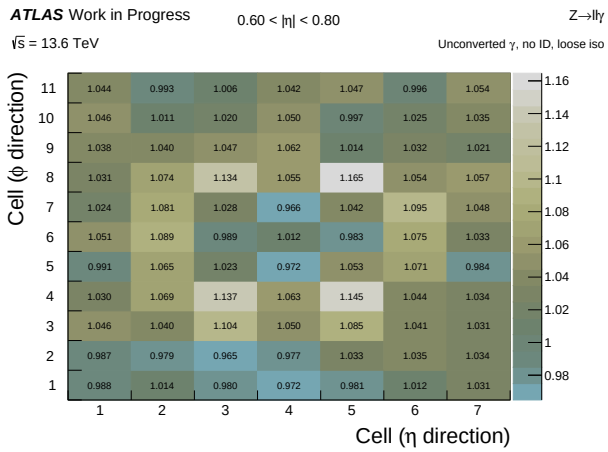
(b) M2 stretch (s_k)

Figure 72: Correction maps, $|\eta| \in [0.40, 0.60)$.

$|\eta| \in [0.60, 0.80)$



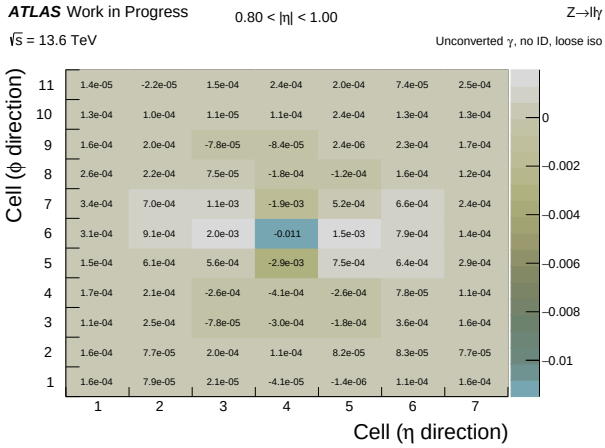
(a) M2 shift (a_k)



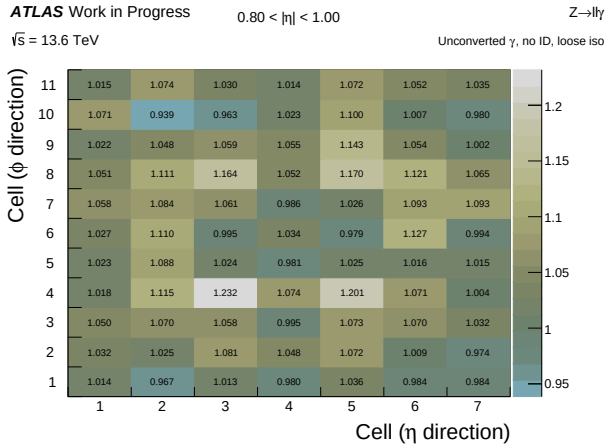
(b) M2 stretch (s_k)

Figure 73: Correction maps, $|\eta| \in [0.60, 0.80)$.

$|\eta| \in [0.80, 1.00)$



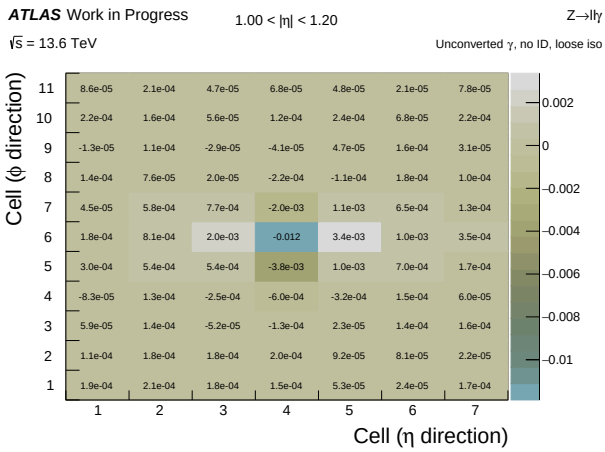
(a) M2 shift (a_k)



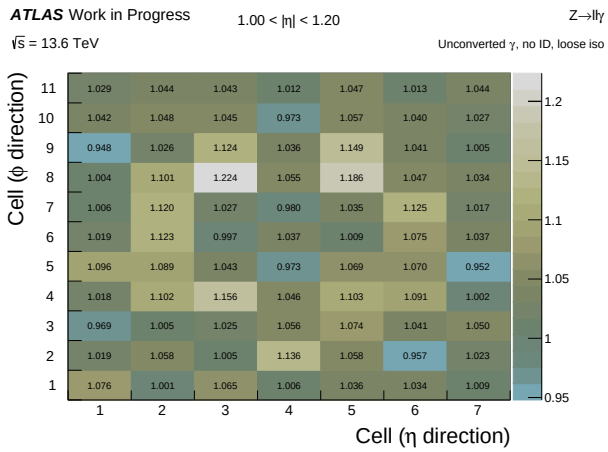
(b) M2 stretch (s_k)

Figure 74: Correction maps, $|\eta| \in [0.80, 1.00)$.

$|\eta| \in [1.00, 1.20)$



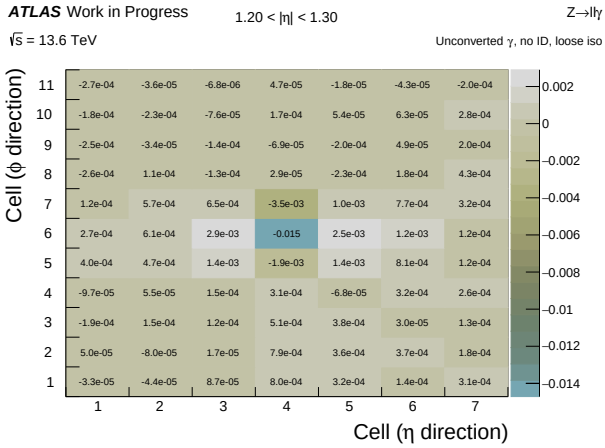
(a) M2 shift (a_k)



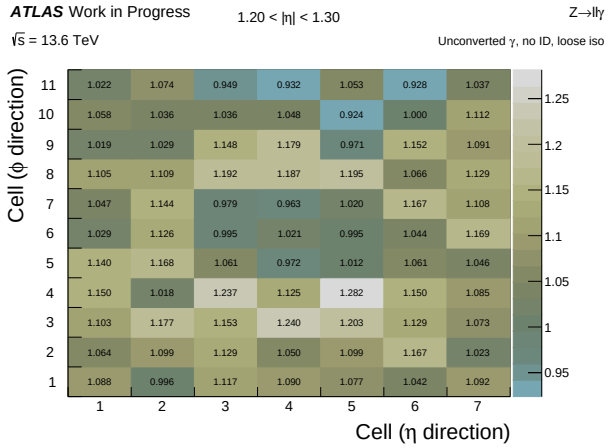
(b) M2 stretch (s_k)

Figure 75: Correction maps, $|\eta| \in [1.00, 1.20)$.

$|\eta| \in [1.20, 1.30)$



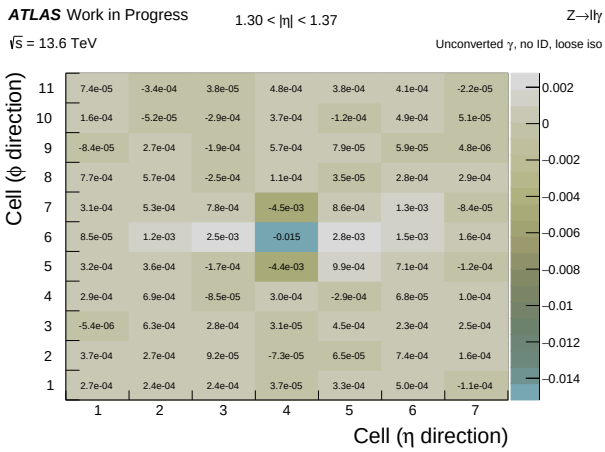
(a) M2 shift (a_k)



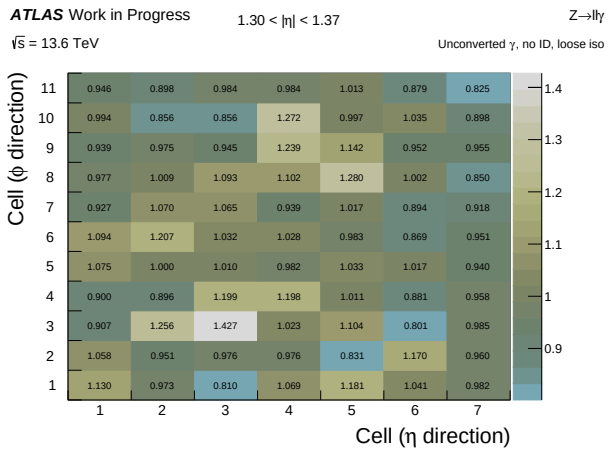
(b) M2 stretch (s_k)

Figure 76: Correction maps, $|\eta| \in [1.20, 1.30)$.

$|\eta| \in [1.30, 1.37)$



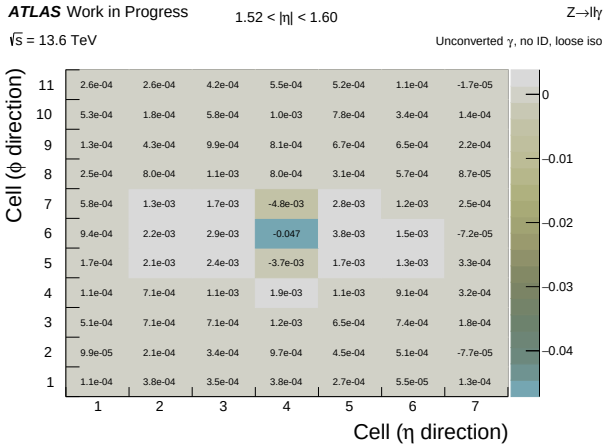
(a) M2 shift (a_k)



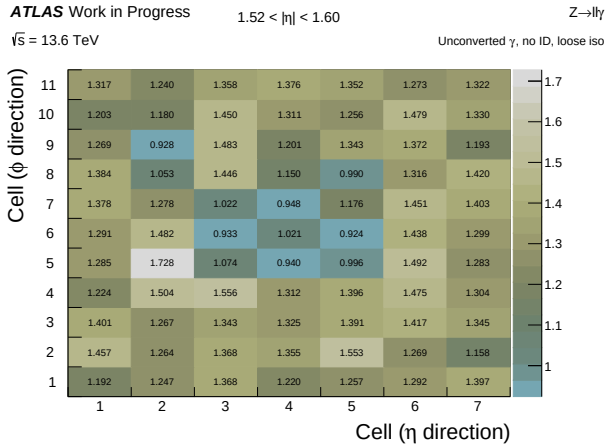
(b) M2 stretch (s_k)

Figure 77: Correction maps, $|\eta| \in [1.30, 1.37)$.

$|\eta| \in [1.52, 1.60)$



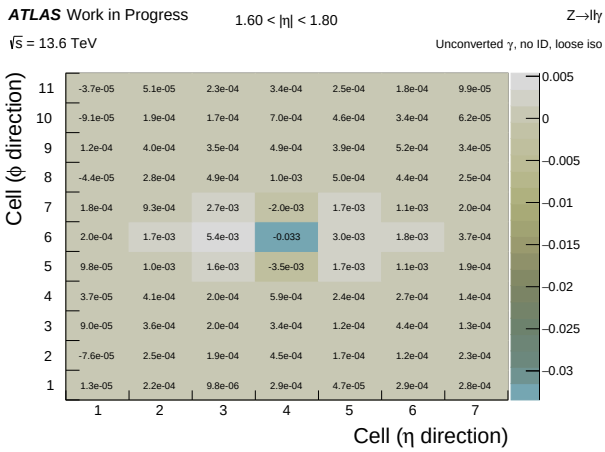
(a) M2 shift (a_k)



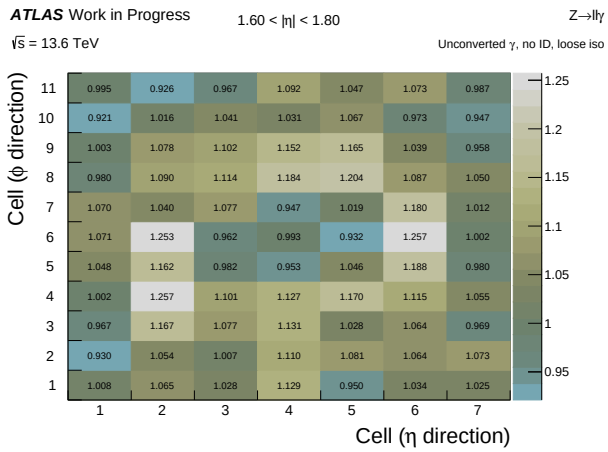
(b) M2 stretch (s_k)

Figure 78: Correction maps, $|\eta| \in [1.52, 1.60)$.

$|\eta| \in [1.60, 1.80)$



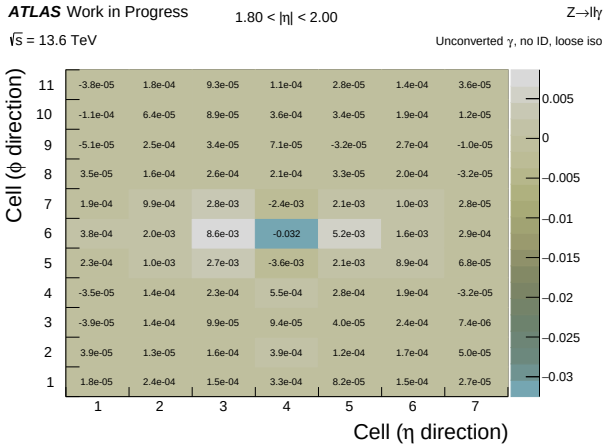
(a) M2 shift (a_k)



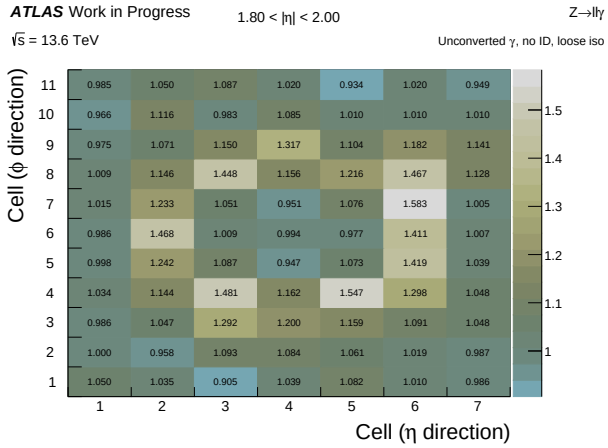
(b) M2 stretch (s_k)

Figure 79: Correction maps, $|\eta| \in [1.60, 1.80)$.

$|\eta| \in [1.80, 2.00)$



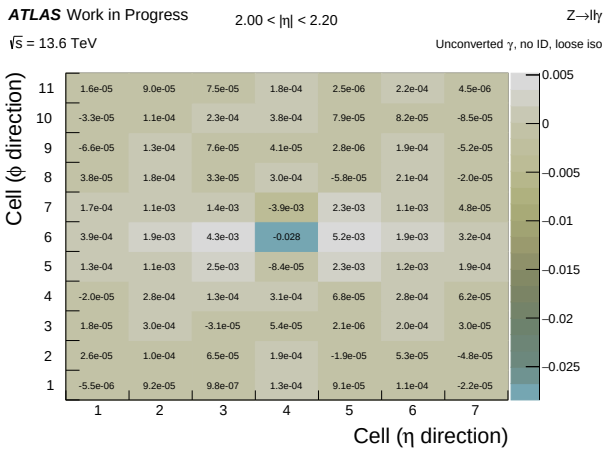
(a) M2 shift (a_k)



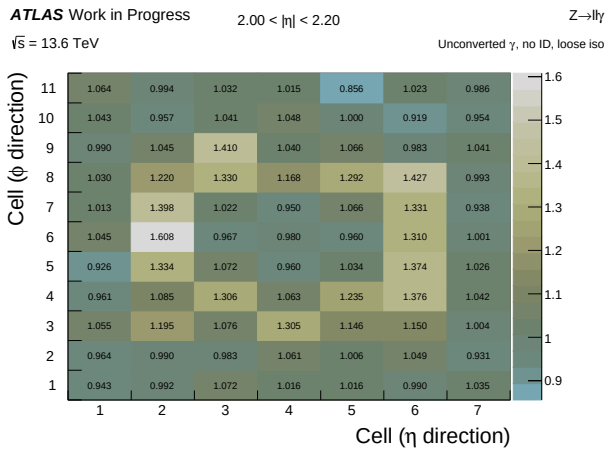
(b) M2 stretch (s_k)

Figure 80: Correction maps, $|\eta| \in [1.80, 2.00)$.

$|\eta| \in [2.00, 2.20)$



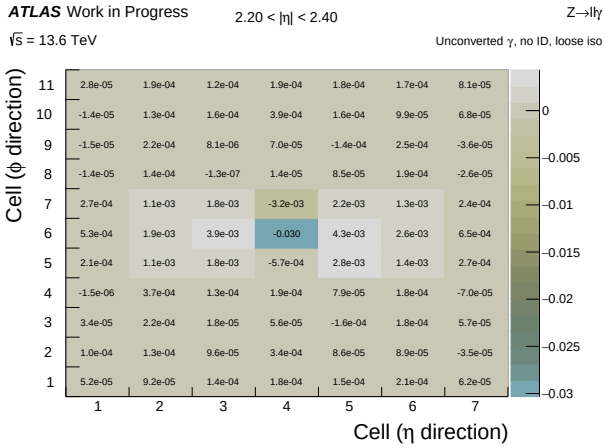
(a) M2 shift (a_k)



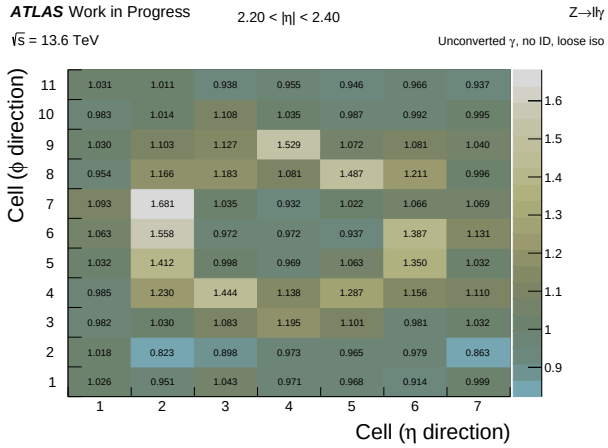
(b) M2 stretch (s_k)

Figure 81: Correction maps, $|\eta| \in [2.00, 2.20)$.

$|\eta| \in [2.20, 2.40)$



(a) M2 shift (a_k)



(b) M2 stretch (s_k)

Figure 82: Correction maps, $|\eta| \in [2.20, 2.40)$.